

LightSpeed 4.X
Functional Interconnects & Block Diagrams



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GE HEALTHCARE . UNLICENSED USE IS STRICTLY PROHIBITED



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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER’S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER’S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N’EST DISPONIBLE QU’EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L’ANGLAIS, C’EST AU CLIENT QU’IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D’INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N’A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L’OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。
GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.



CAUTION



ATTENTION

LITHIUM BATTERY CAUTIONARY STATEMENTS

Risk of Explosion

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.

Danger d’Explosion

Il y a danger d’explosion s’il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d’un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GE Healthcare iTrak process to report all omissions, errors, and defects in this publication.

Revision History

Rev	Date	Reason for change
4	03/09/07	Added NGPDU information, to support some CT and CT/PET systems shipped in 2006 or later.
3	05/10/05	PQR 13024644: Updated drawing 2267770SCH and added drawing 2267713SCH.
2	08/17/04	PQR 13012919: Added Section 5.0 — Table Cradle Drive (Vertical & Longitudinal Drives) - 2123540 to Chapter 4.
1	06/01/04	Initial Release (fomerly part of Dir 2341960-100)

Table of Contents

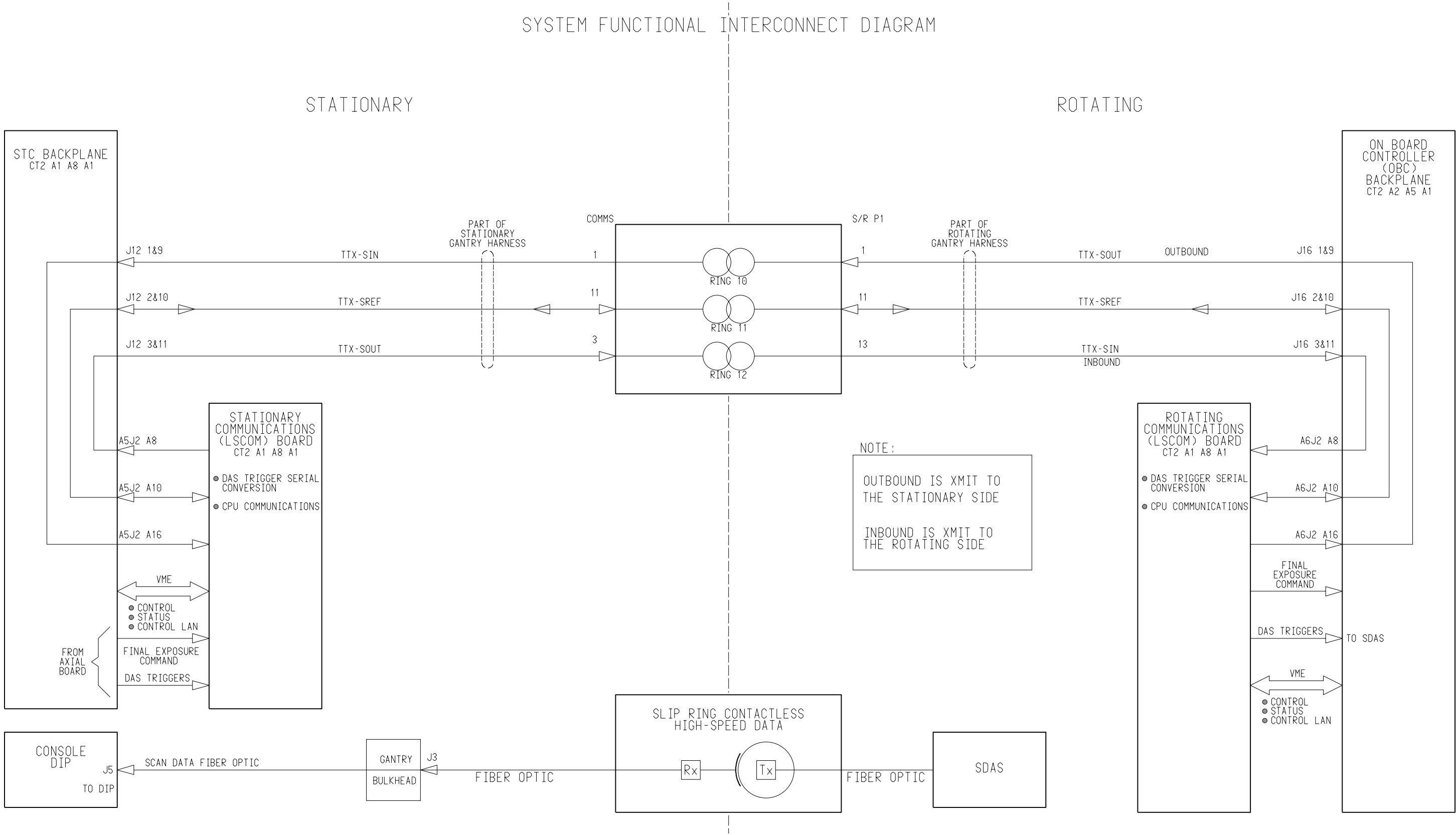
Chapter 1		
System Communications	7	
Section 1.0 — Gantry Slip Ring Communications - 2267732	8	
Section 2.0 — Rotating Controller Interface Bus (RCIB) - 2305060	9	
Section 3.0 — Final Scan Control / Intercom / AutoVoice - 2151953	10	
 Chapter 2		
X-Ray Generation.....	13	
Section 1.0 — Rotor Control - 2267317	14	
Section 2.0 — Exposure Enable - 2267735	15	
Section 3.0 — Rotating HVON Sense - 2267736	16	
Section 4.0 — Filament Select and mA Loop - 2267737	17	
Section 5.0 — Tube ID, Cooling and Pressure Sense - 2267738	18	
Section 6.0 — Back-Up Timer - 2267739	19	
Section 7.0 — kV Loop - 2267740.....	20	
Section 8.0 — BU Contactor / Exposure Interlocks - 2267741	22	
Section 9.0 — X-Ray Collimation and Filtration - 2267744.....	25	
Section 10.0 — Final Exposure Command - 2267746.....	26	
 Chapter 3		
Data Acquisition	27	
Section 1.0 — DAS Power Supplies - 2302185	28	
Section 2.0 — Detector Heater Control Board - 2302184	29	
 Chapter 4		
Patient Positioning	31	
Section 1.0 — Patient Alignment Lights - 2126249	32	
Section 2.0 — Table Gantry Sync - 2267745	33	
Section 3.0 — Table Elevation Drive - 2267754	34	
Section 4.0 — Gantry Tilt Function - 2267756.....	35	
Section 5.0 — Table Cradle Drive (Vertical & Longitudinal Drives) - 2123540.....	36	
 Chapter 5		
Axial Control	39	
Section 1.0 — Axial Servo Control Loop - 2267757	40	
Section 2.0 — Axial Drive Power Control - 2267784	41	
 Chapter 6		
Operator I/O	43	
Section 1.0 — Gantry Operator Interface - 2267760	44	
Section 2.0 — Site X-Ray ON Light - 2267761	45	
Section 3.0 — Site System ON Light.....	47	
Section 4.0 — RX Tilt & Console Gantry Reset - 2267765.....	48	
 Chapter 7		
System Monitoring.....	49	
Section 1.0 — Mains Undervoltage - 2267731 (Not Applicable to NGPDU)	50	
Section 2.0 — Gantry Temp. Sensors & Power Supply Monitors - 2267766.....	51	
 Chapter 8		
System Power Control.....	53	
Section 1.0 — HVDC Bus Control - 2267769	54	
Section 2.0 — Gantry 120VAC Control - 2267770.....	57	
Section 3.0 — E-Stop and Drives Reset - 2267773	59	
Section 4.0 — Gantry Stationary Power and Ground - 2267713	61	
 Chapter 9		
Image Generation (Global Console - Linux).....	65	
Section 1.0 — Scan Data Save	66	
Section 2.0 — Data Restore.....	67	
Section 3.0 — Pre-Processing, Filtered Back Projection & Post Processing	68	
Section 4.0 — Image Transfer	69	
 Chapter 10		
Image Generation (Global Console - Octane2).....	71	
Section 1.0 — Scan Data Save	72	
Section 2.0 — Data Restore.....	73	

Section 3.0 — Pre-Processing, Filtered Back Projection & Post Processing.....	74
Section 4.0 — Image Transfer	75

Chapter 1

System Communications

Section 1.0 – Gantry Slip Ring Communications - 2267732



Section 2.0 – Rotating Controller Interface Bus (RCIB) - 2305060

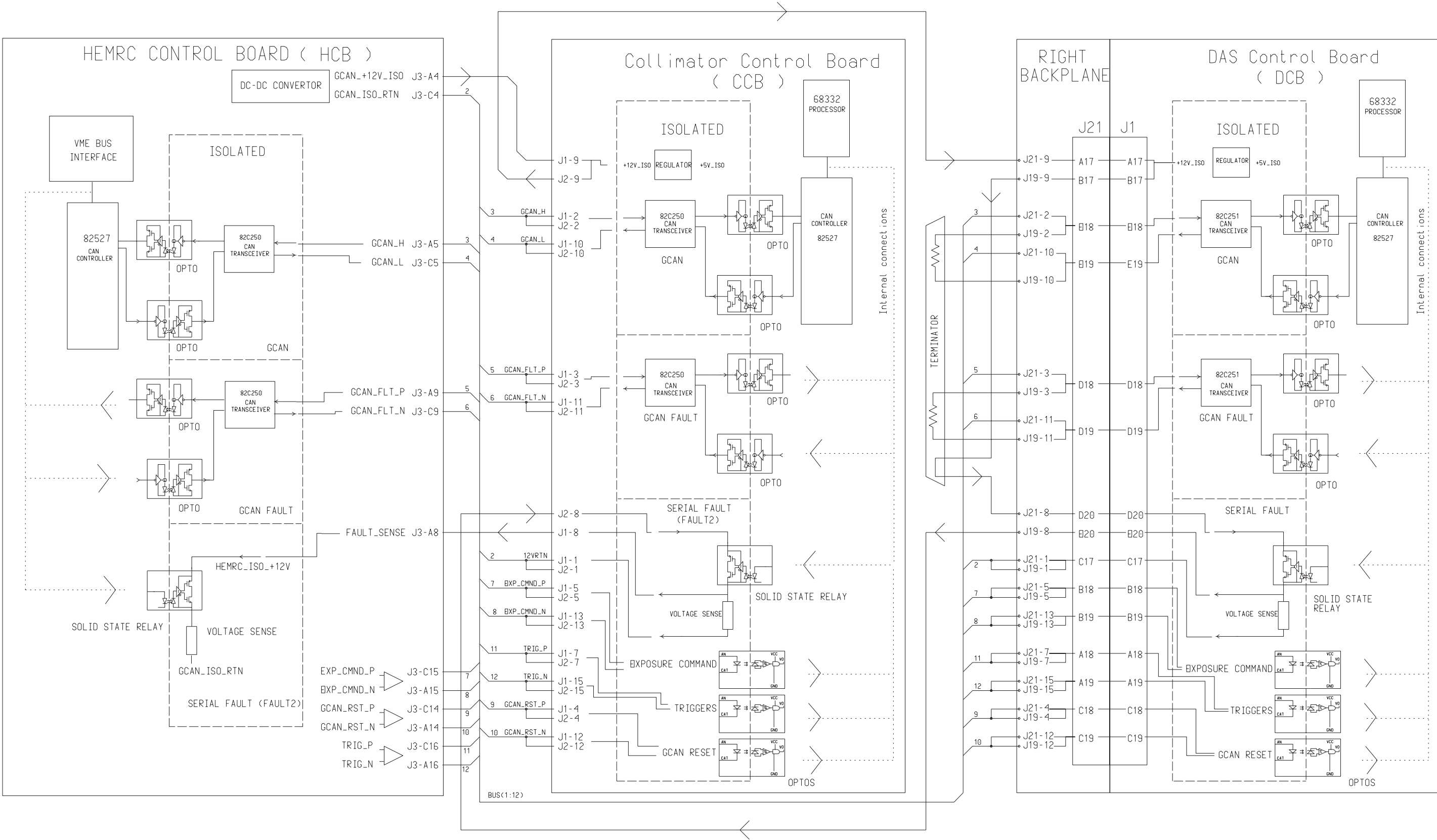
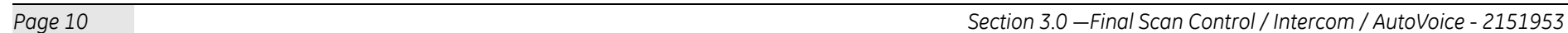


Figure 1-2 2305060 Sheet 1, Rev. 0

Figure 1-3 2151953 Sheet 1, Rev 1



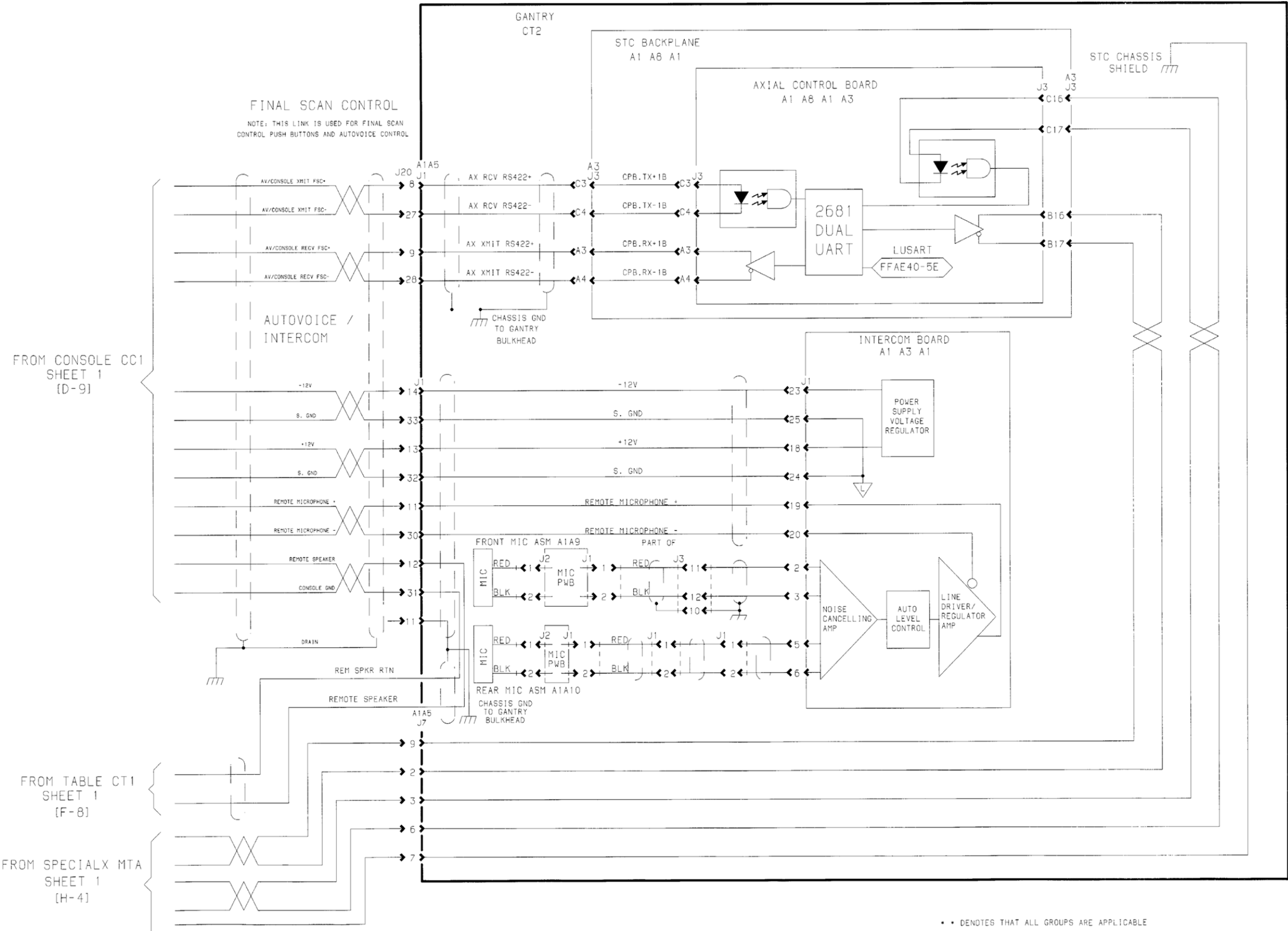


Figure 1-4 2151953 Sheet 2, Rev 1

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Chapter 2

X-Ray Generation

Section 1.0 – Rotor Control - 2267317

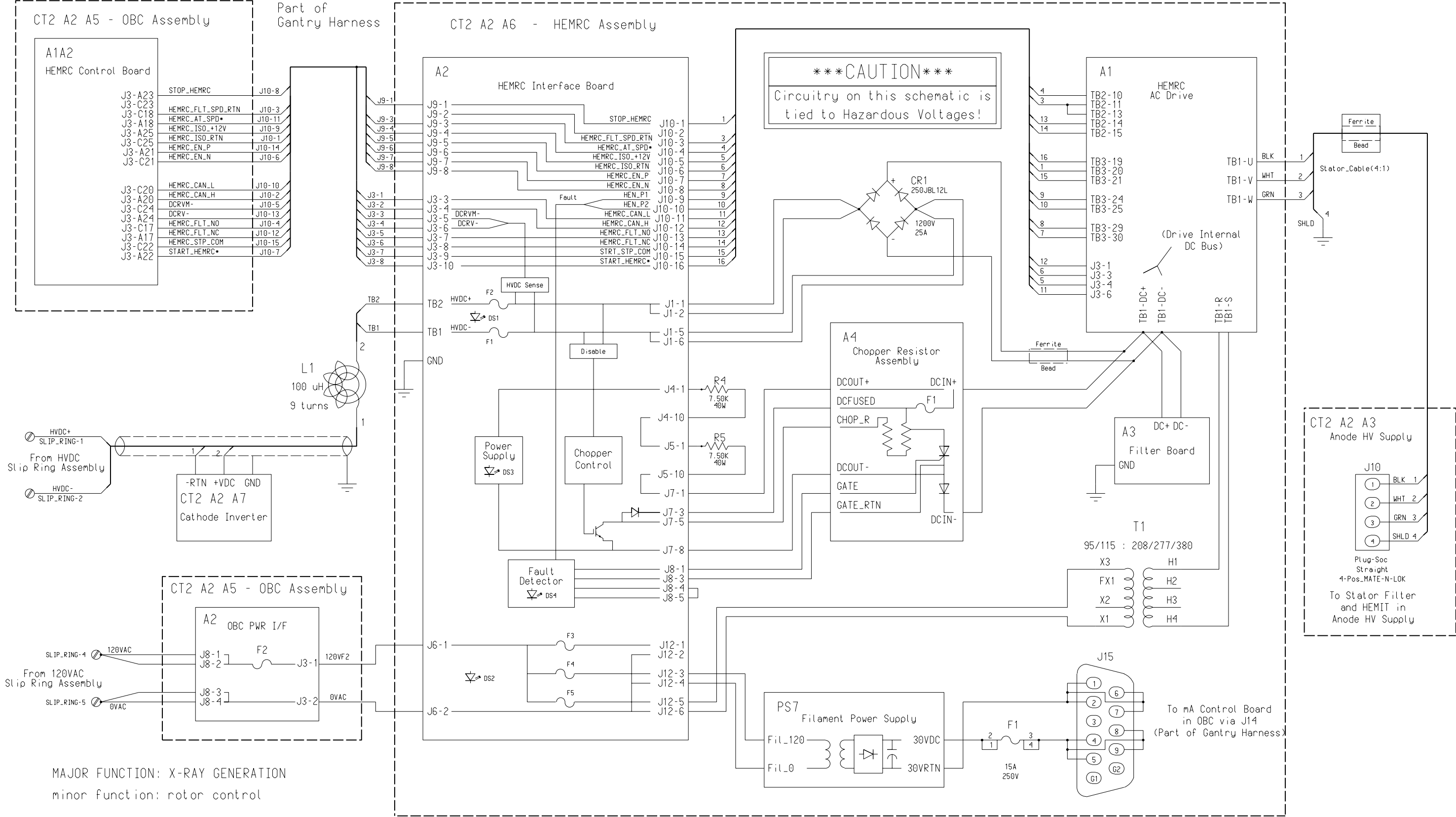


Figure 2-1 2267317 Sheet 1, Rev. 0

Section 2.0 – Exposure Enable - 2267735

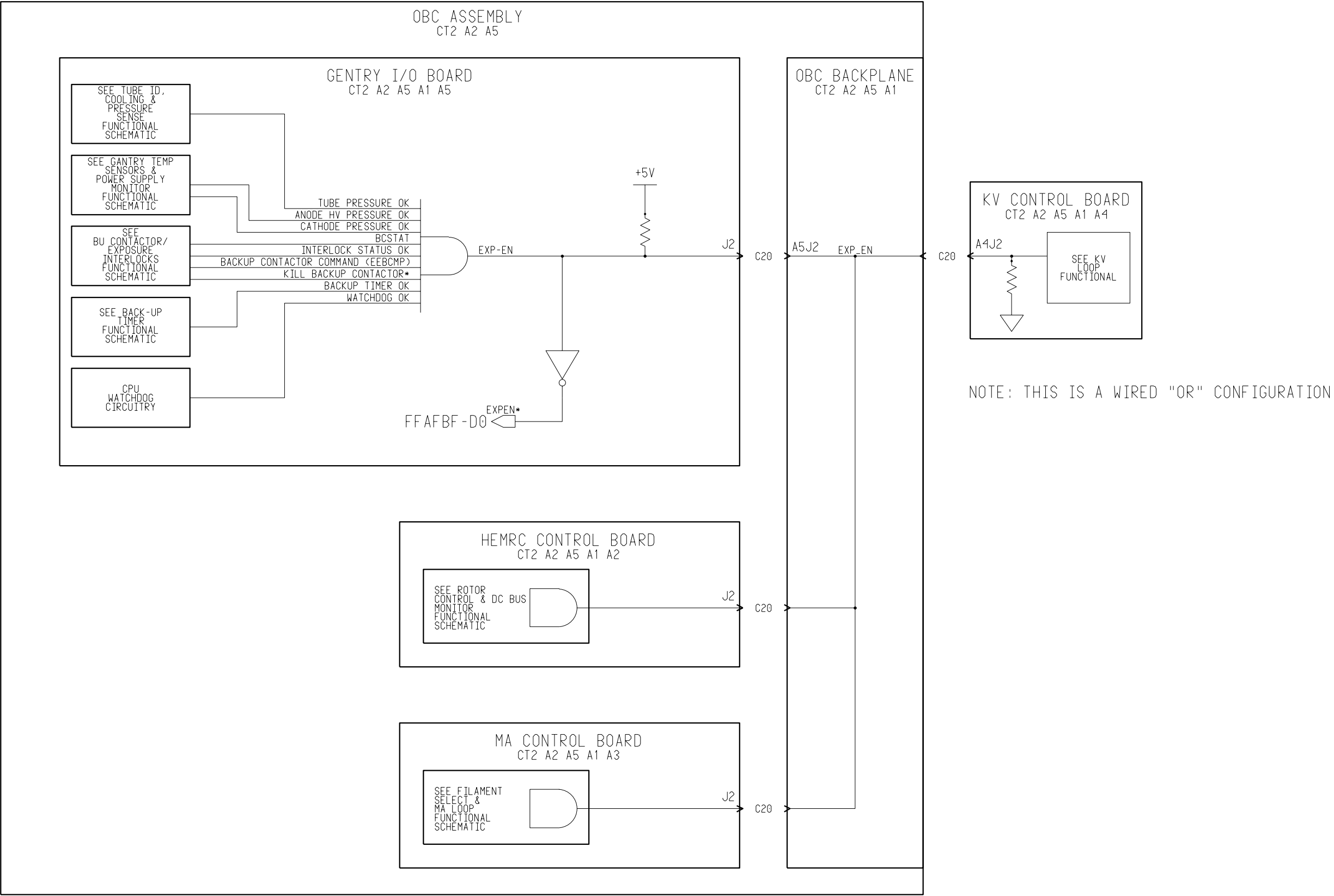


Figure 2-2 2267735 Sheet 1, Rev. 0

Section 3.0 – Rotating HVON Sense - 2267736

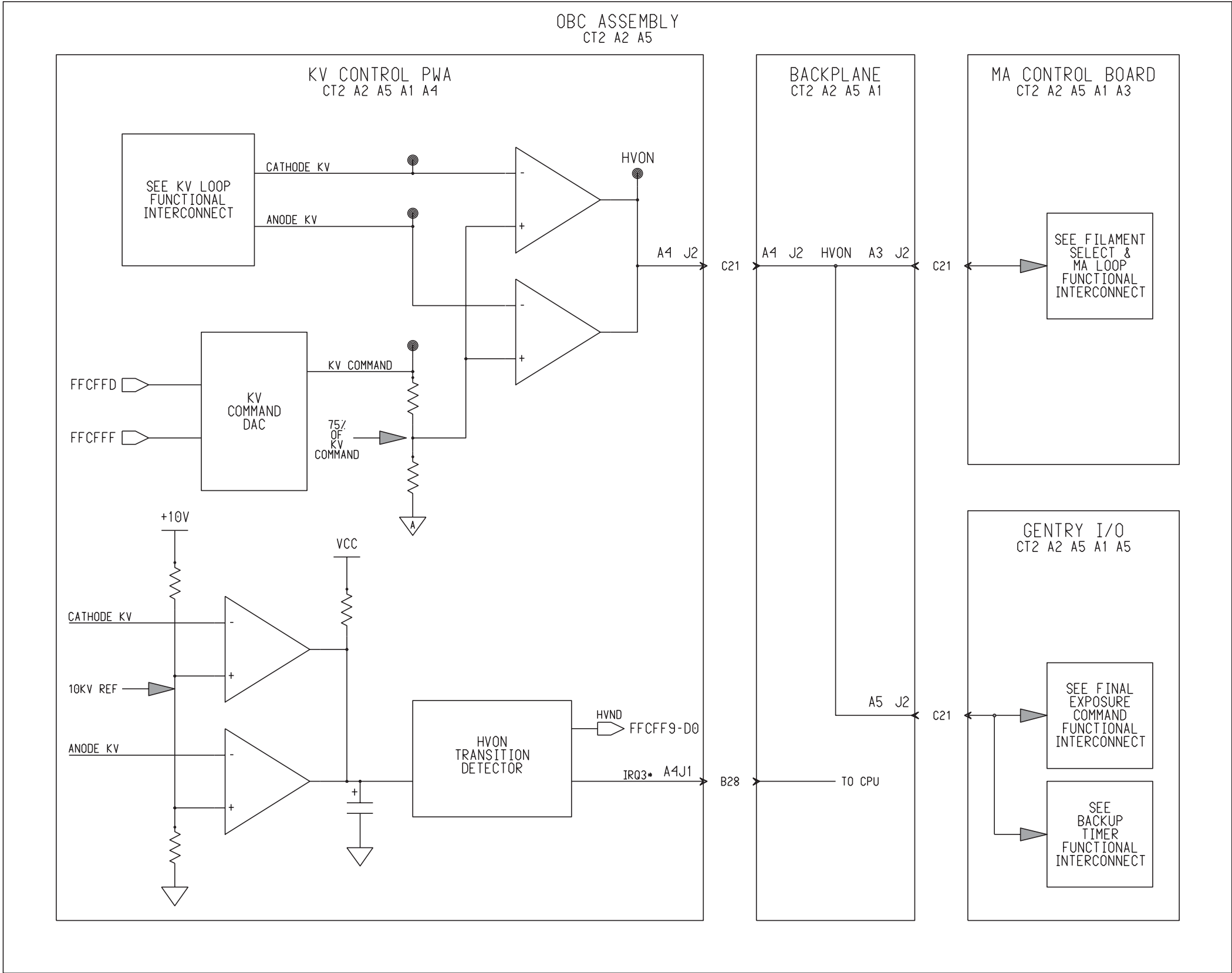


Figure 2-3 2267736 Sheet 1, Rev. 0

Section 4.0 – Filament Select and mA Loop - 2267737

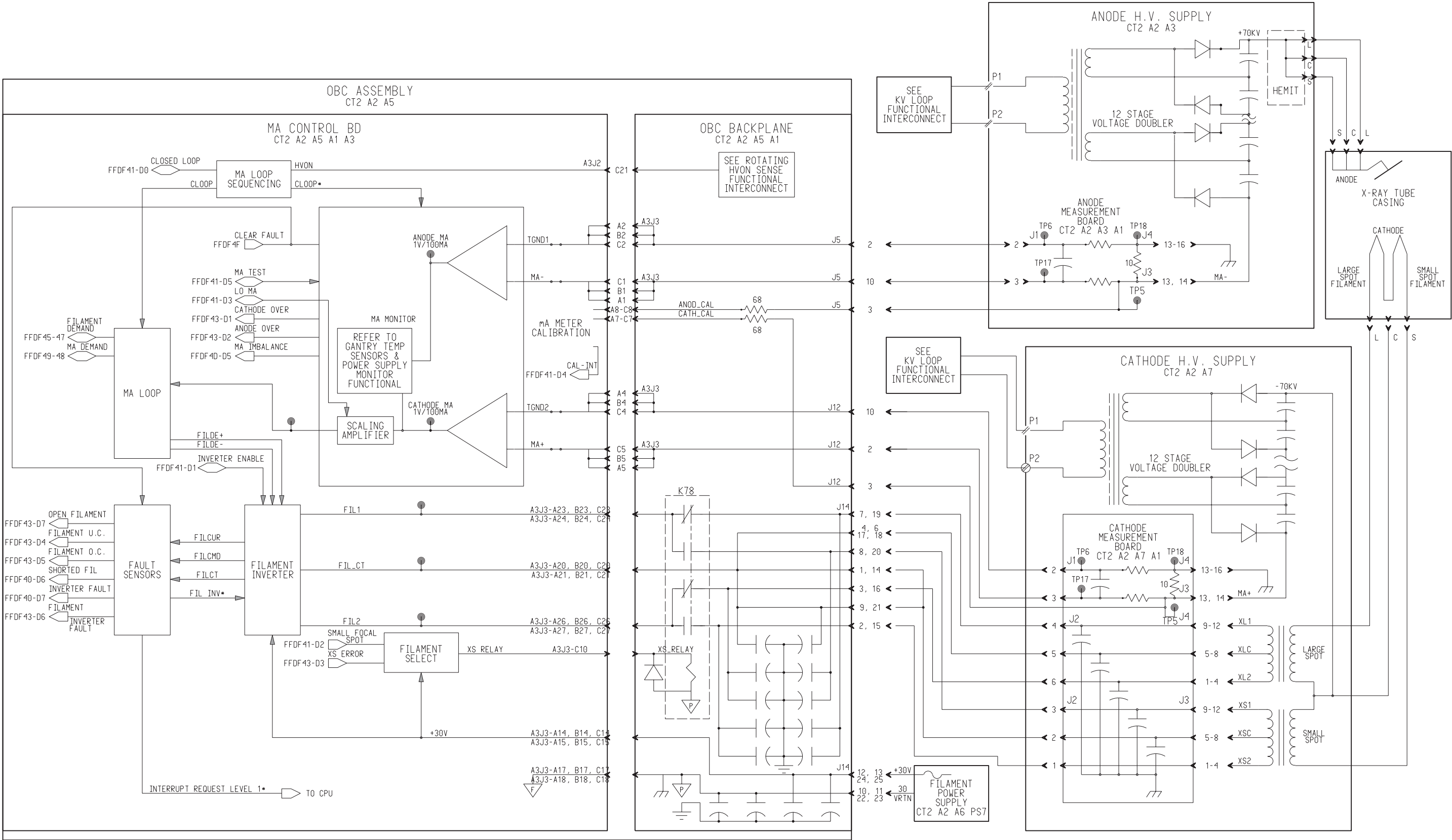


Figure 2-4 2267737 Sheet 1, Rev. 0

Section 5.0 – Tube ID, Cooling and Pressure Sense - 2267738

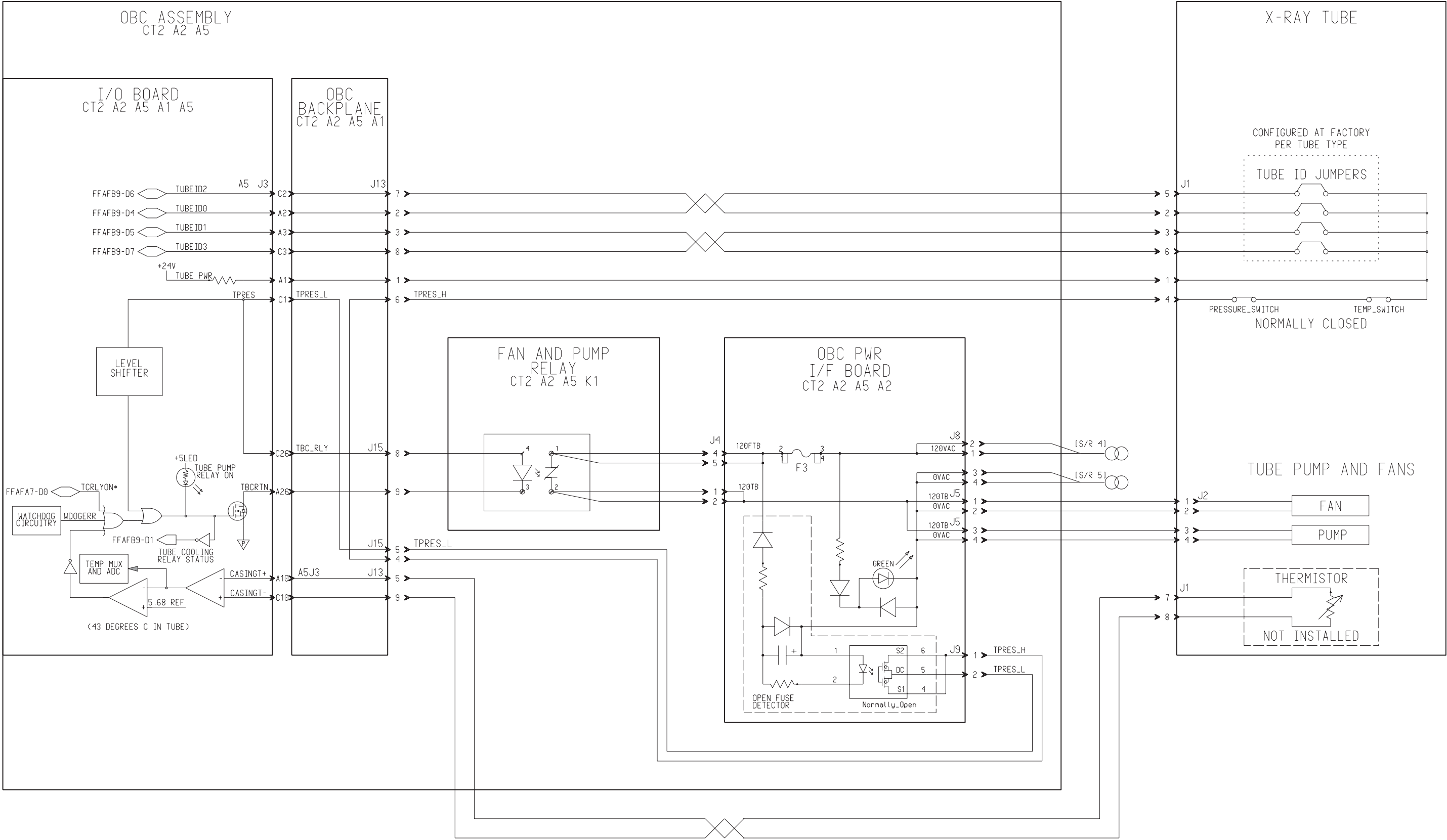


Figure 2-5 2267738 Sheet 1, Rev. 0

Section 6.0 – Back-Up Timer - 2267739

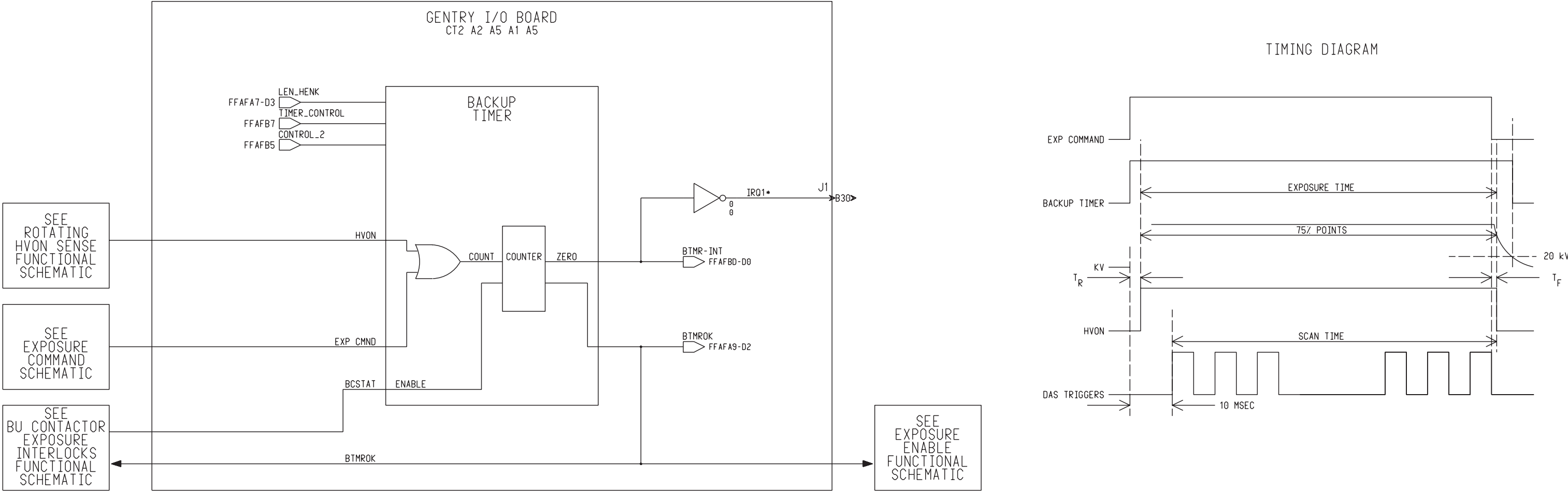
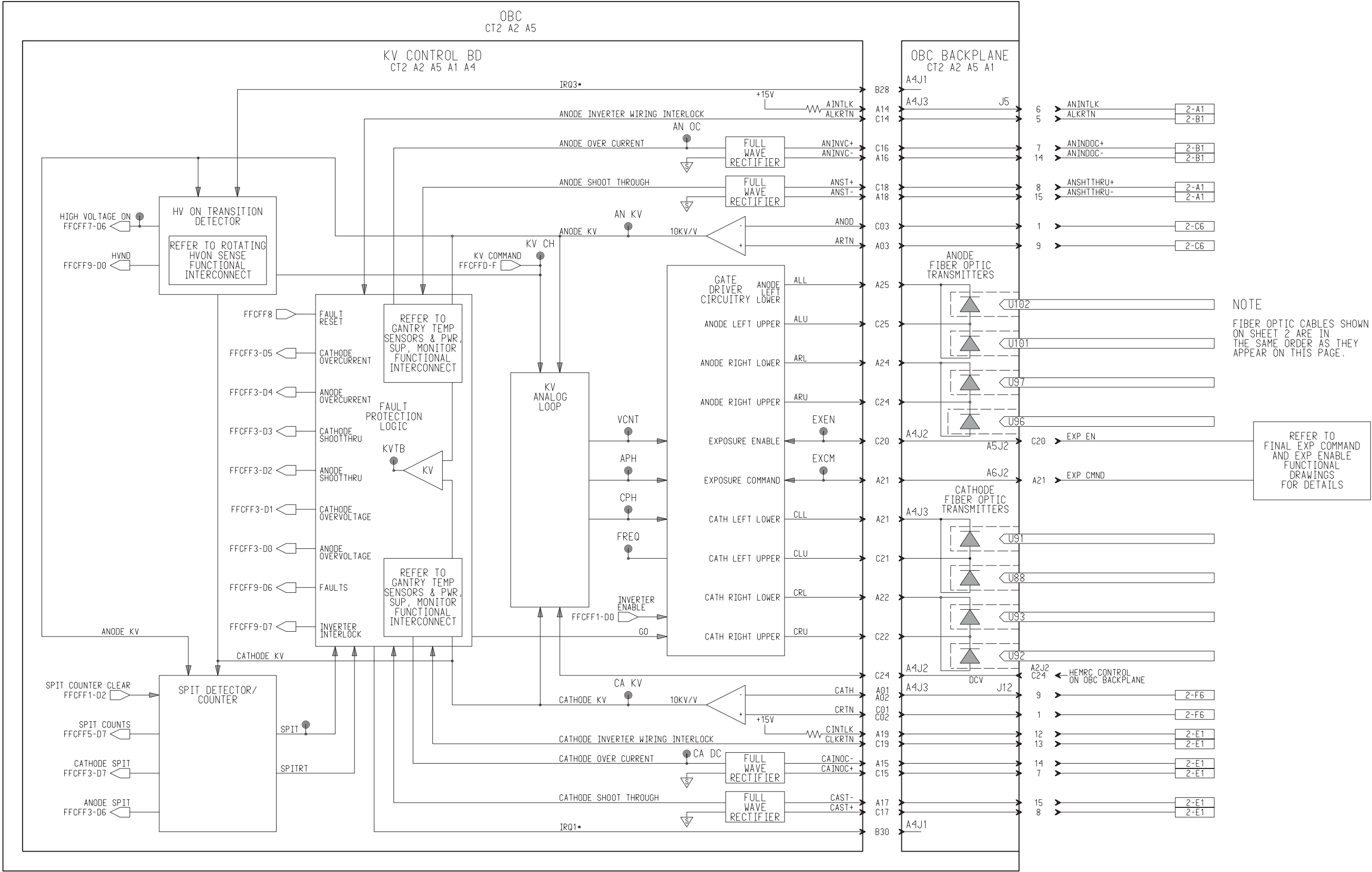


Figure 2-6 2267739 Sheet 1, Rev. 0

Section 7.0 – kV Loop - 2267740



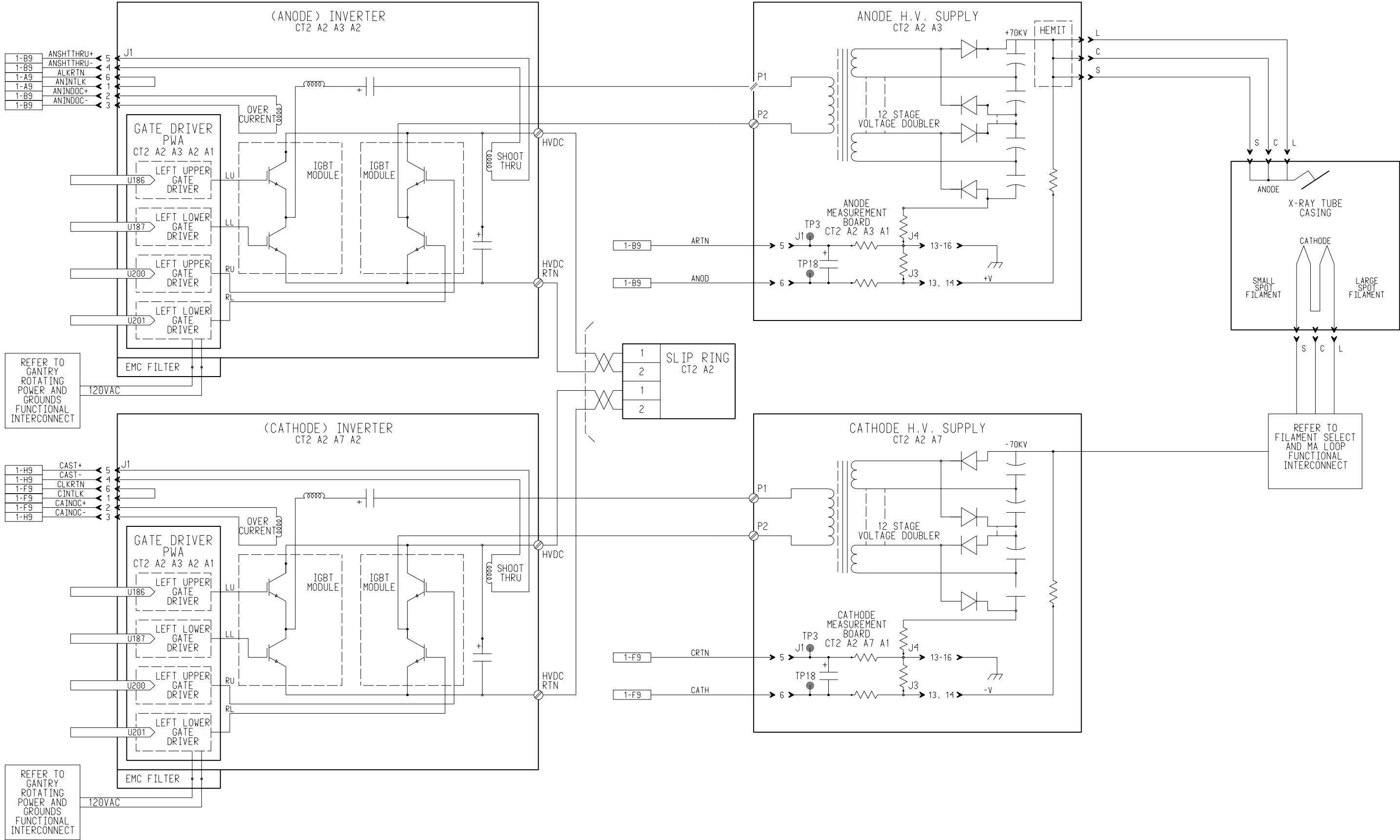


Figure 2-8 2267740 Sheet 2, Rev. 0

Section 8.0 – BU Contactor / Exposure Interlocks - 2267741

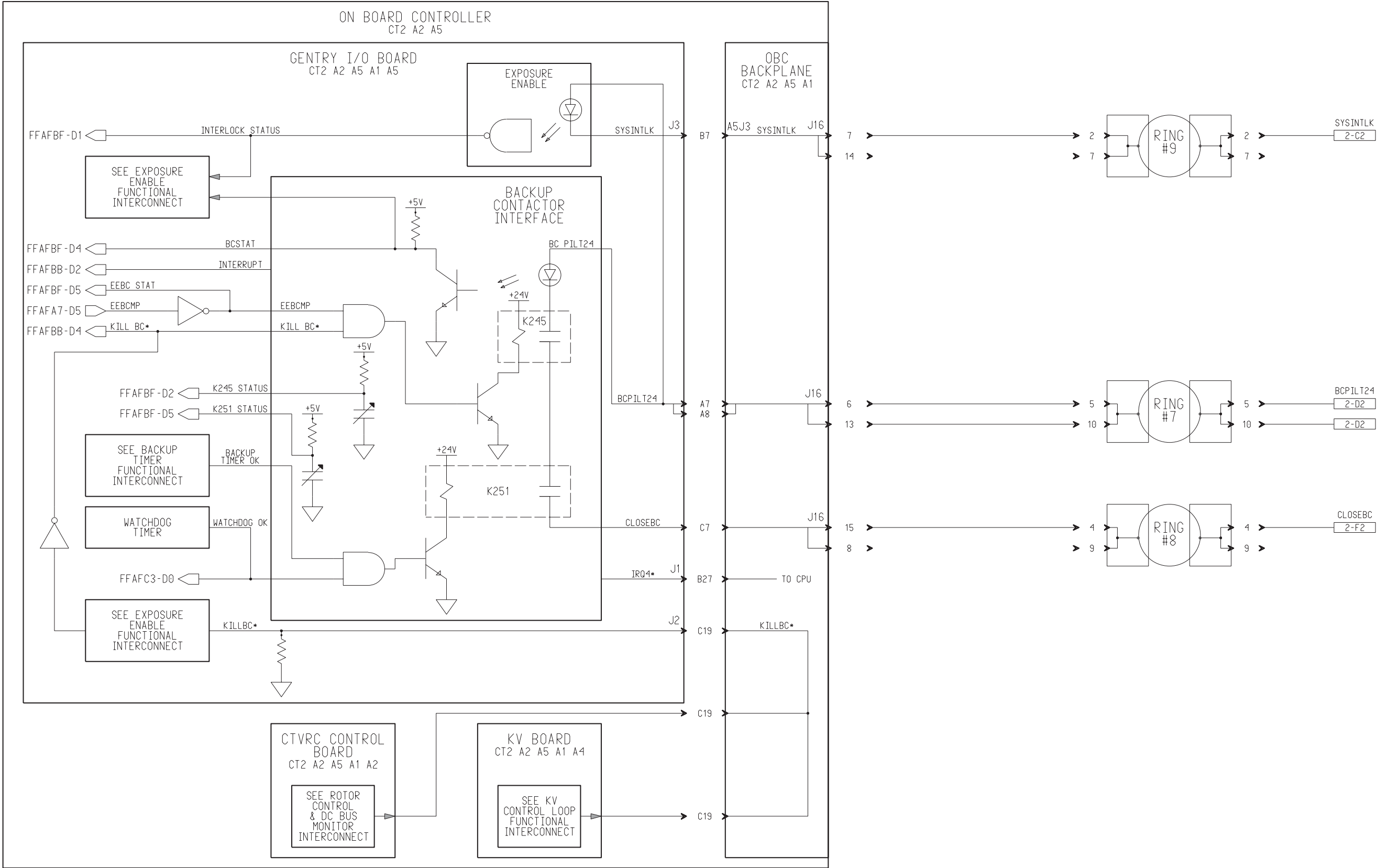


Figure 2-9 2267741 Sheet 1, Rev. 0

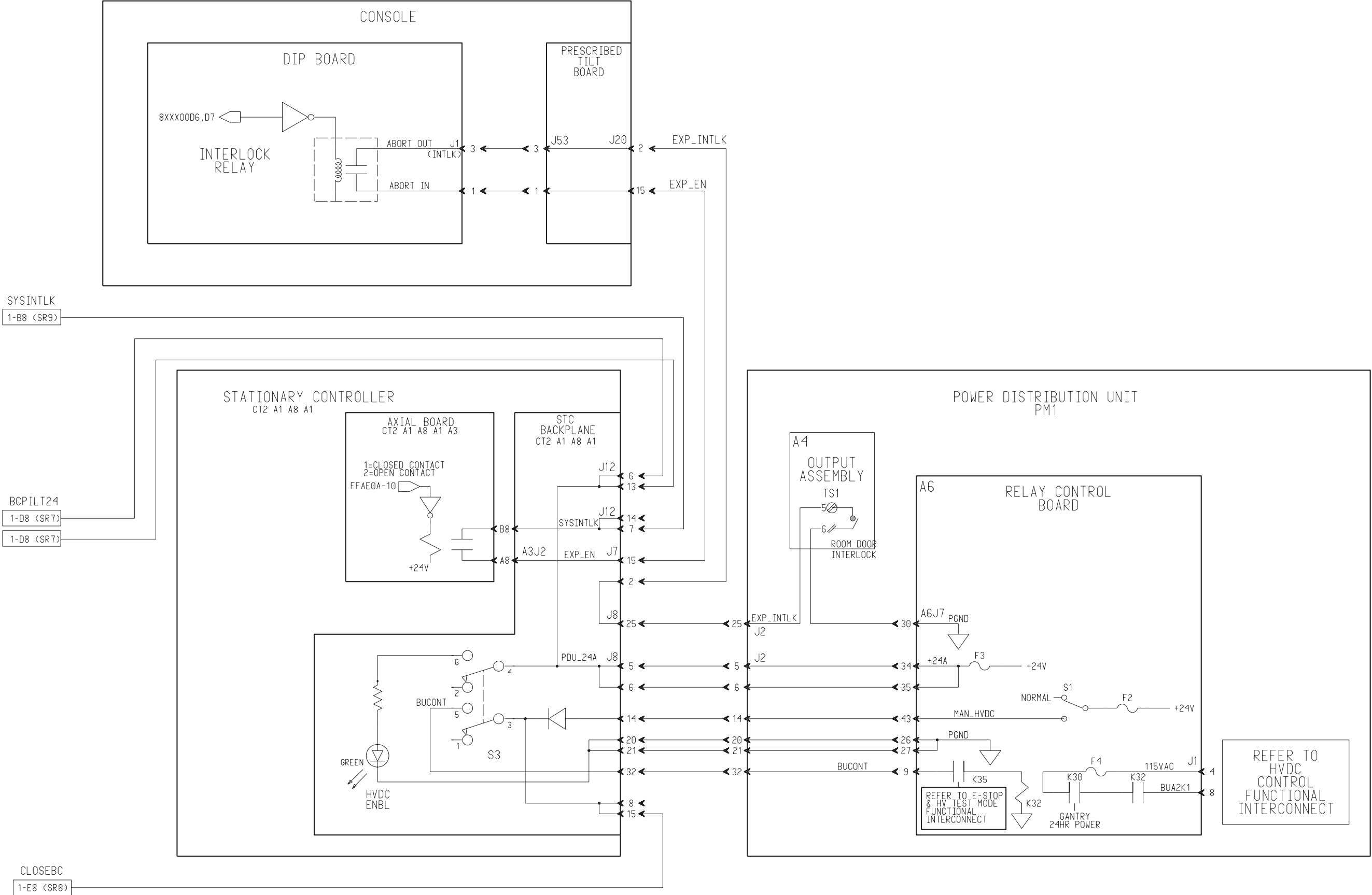


Figure 2-10 2267741 Sheet 2, Rev. 0 (CPDU shown; see Figure 2-11 for NGPDU)

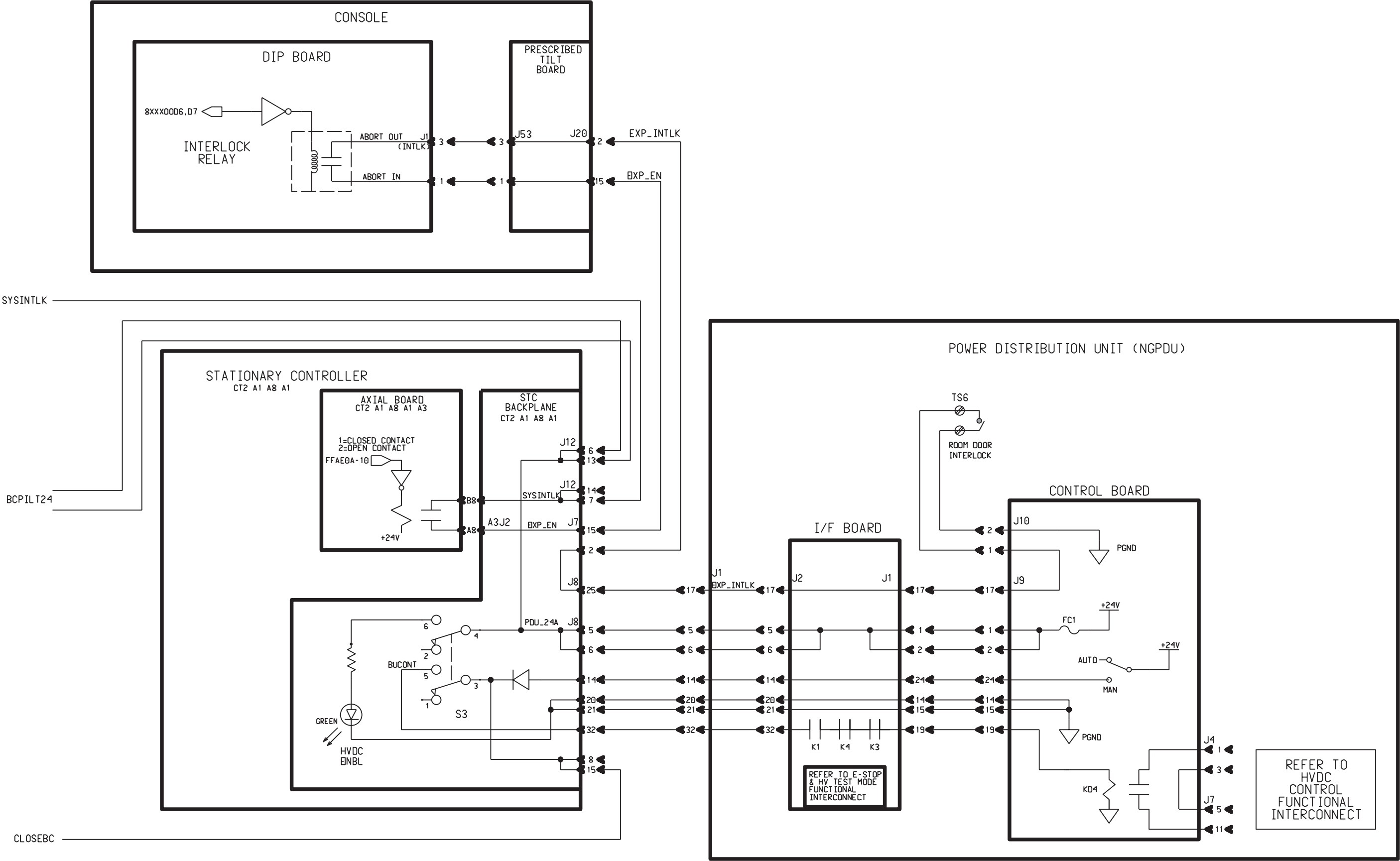


Figure 2-11 BU Contactor / Exposure Interlocks (NGPDU shown; see Figure 2-10 for CPDU)

Section 9.0 – X-Ray Collimation and Filtration - 2267744

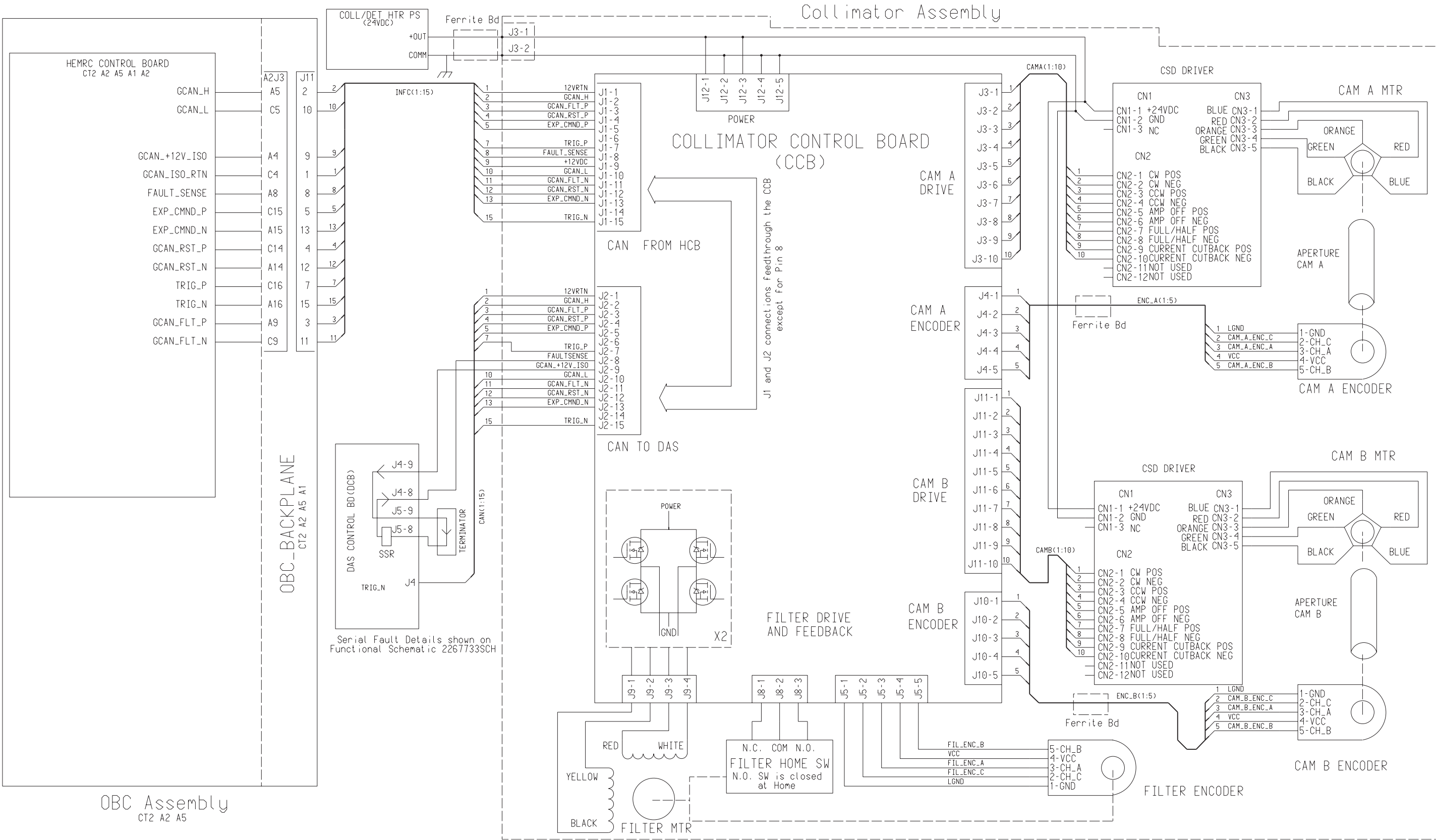


Figure 2-12 2267744 Sheet 1, Rev. 0

Section 10.0 — Final Exposure Command - 2267746

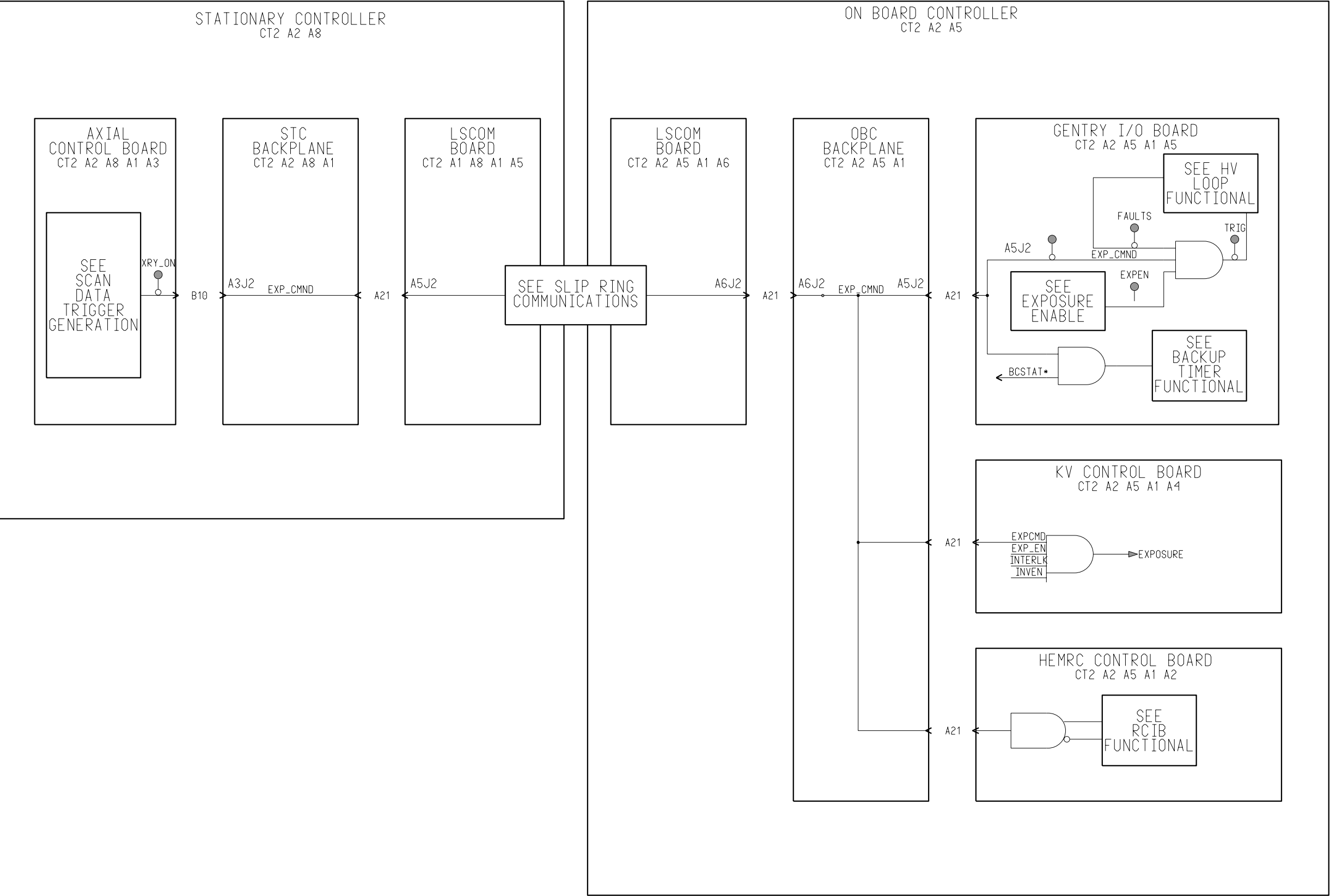


Figure 2-13 2267746 Sheet 1, Rev. 0

Chapter 3

Data Acquisition

Section 1.0 – DAS Power Supplies - 2302185

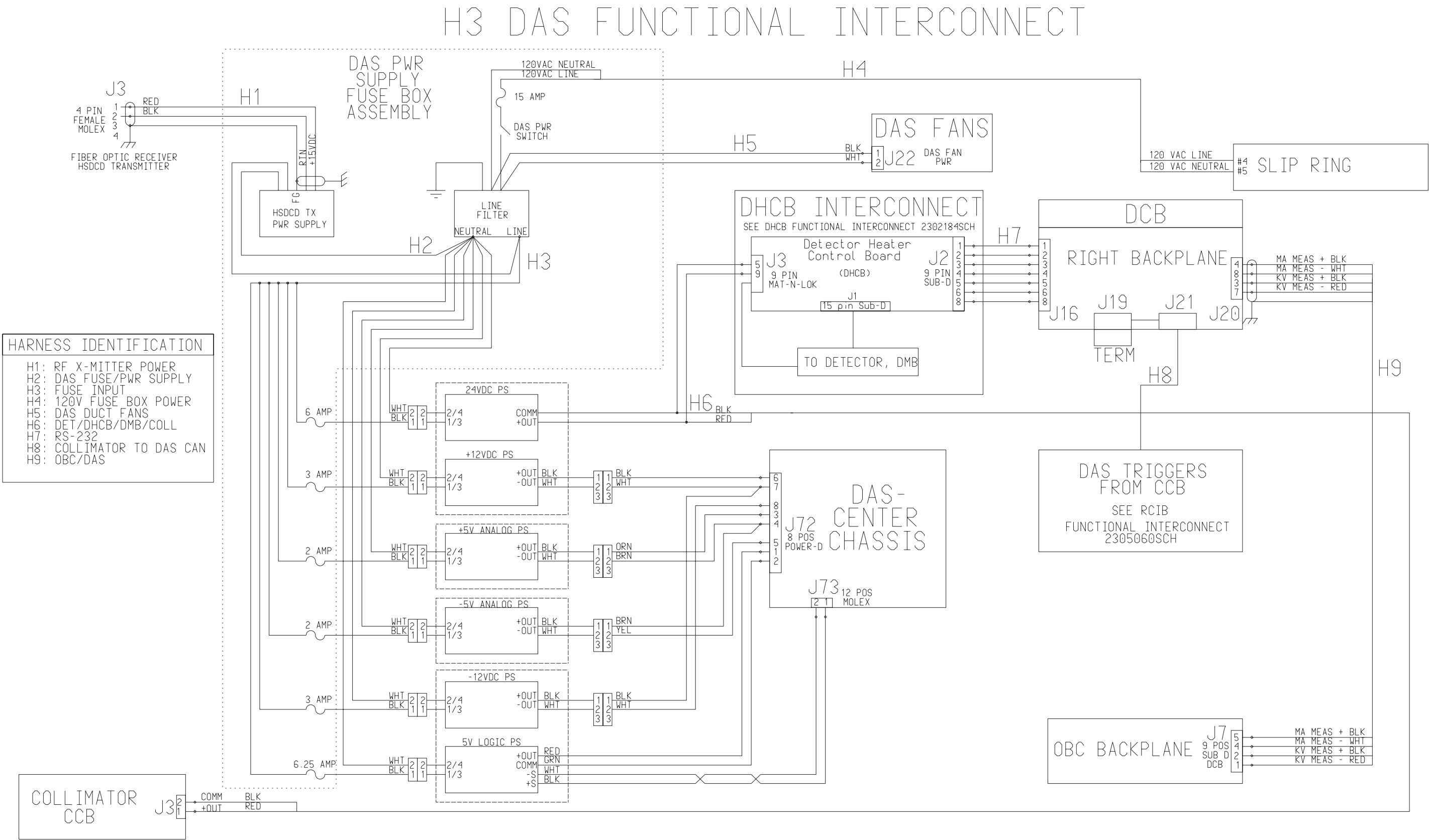


Figure 3-1 2302185 Sheet 1, Rev. 0

Section 2.0 – Detector Heater Control Board - 2302184

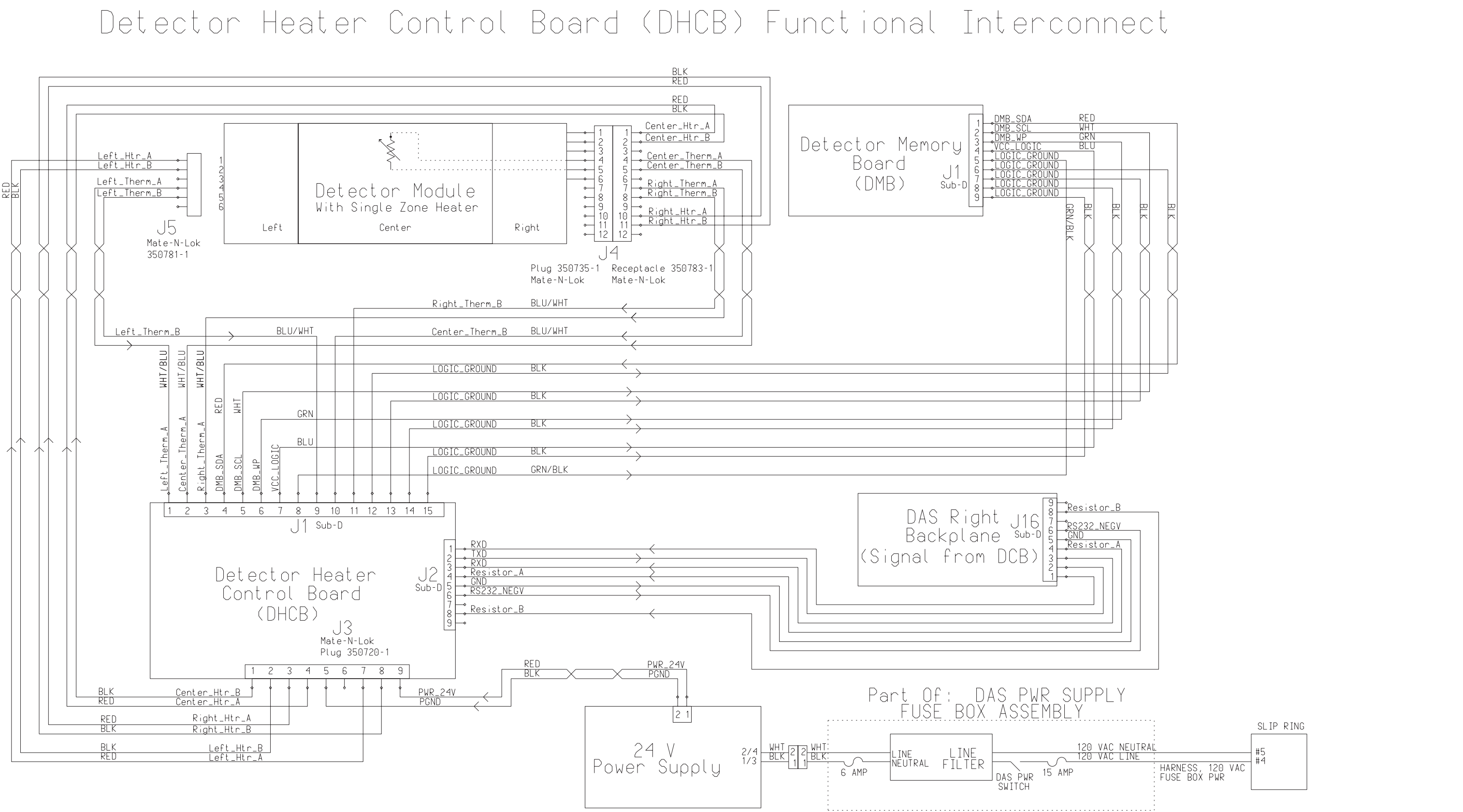


Figure 3-2 2302184 Sheet 1, Rev. 0

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Chapter 4

Patient Positioning

Section 1.0 – Patient Alignment Lights - 2126249

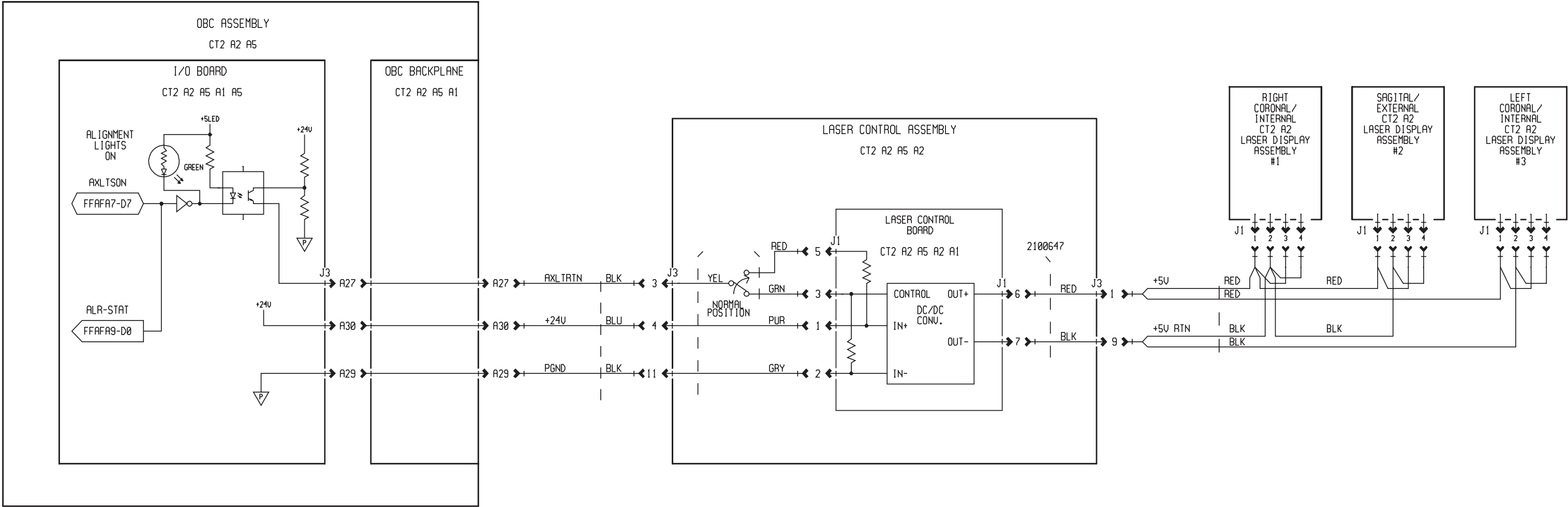
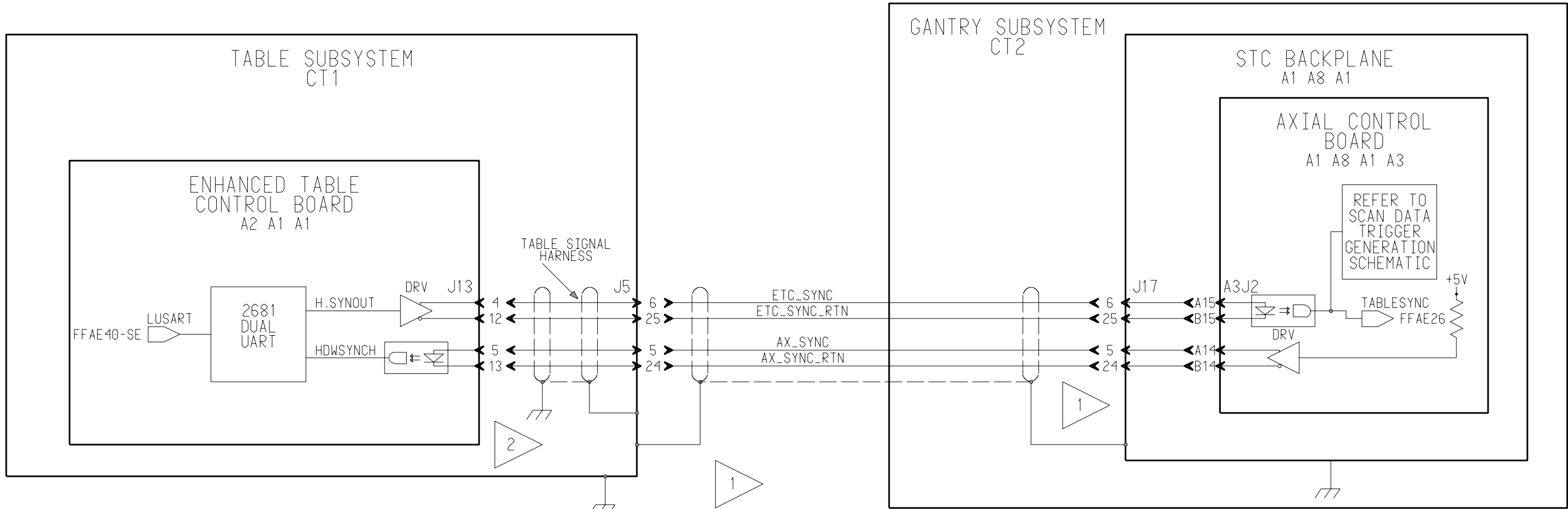


Figure 4-1 2126249 Sheet 1, Rev 0

Section 2.0 – Table Gantry Sync - 2267745



- NOTES
- 1 CABLE SHIELD TIED THROUGH CONNECTOR BACKSHELL TO CHASSIS GROUND.
- 2 CABLE SHIELD TIE WRAPPED TO ETC MOUNTING PLATE.
- 3 ADDRESS

NAME	NAME
ADDRESS	ADDRESS

 INDICATES OUTPUT FROM OR INPUT TO FIRMWARE.

Figure 4-2 2267745 Sheet 1, Rev. 0

ELEVATION

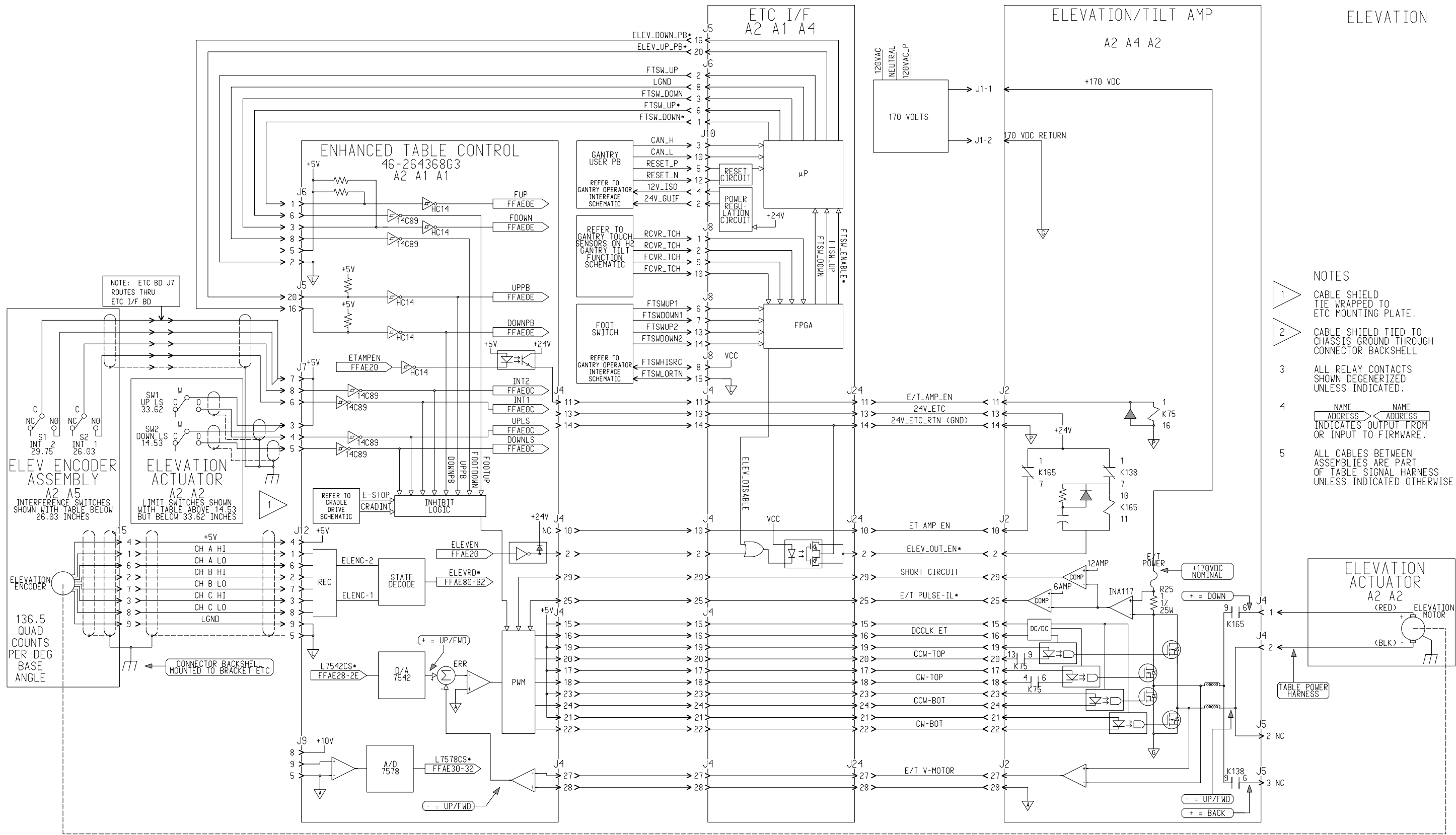


Figure 4-3 2267754 Sheet 1, Rev. 0

Section 4.0 – Gantry Tilt Function - 2267756

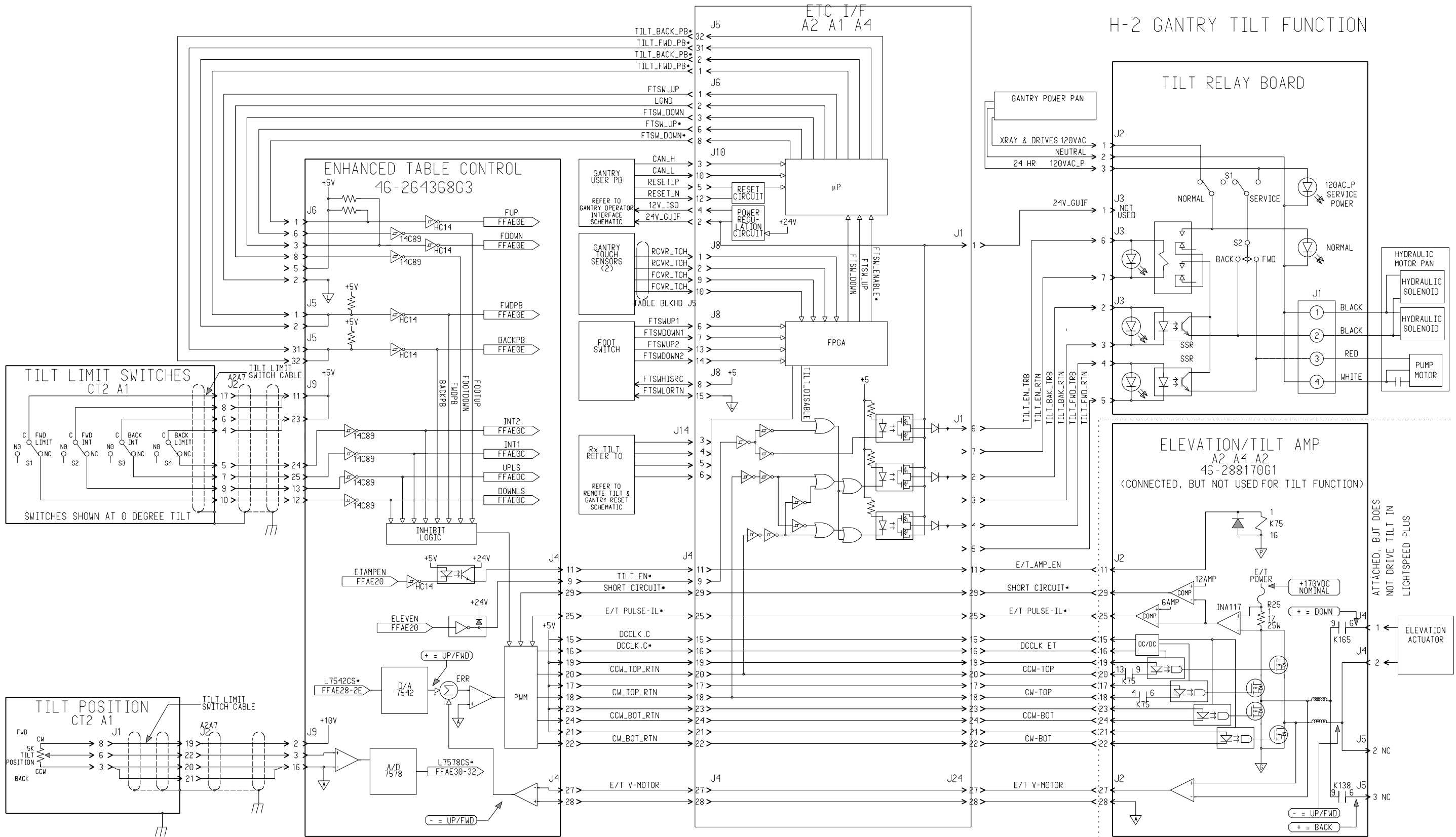
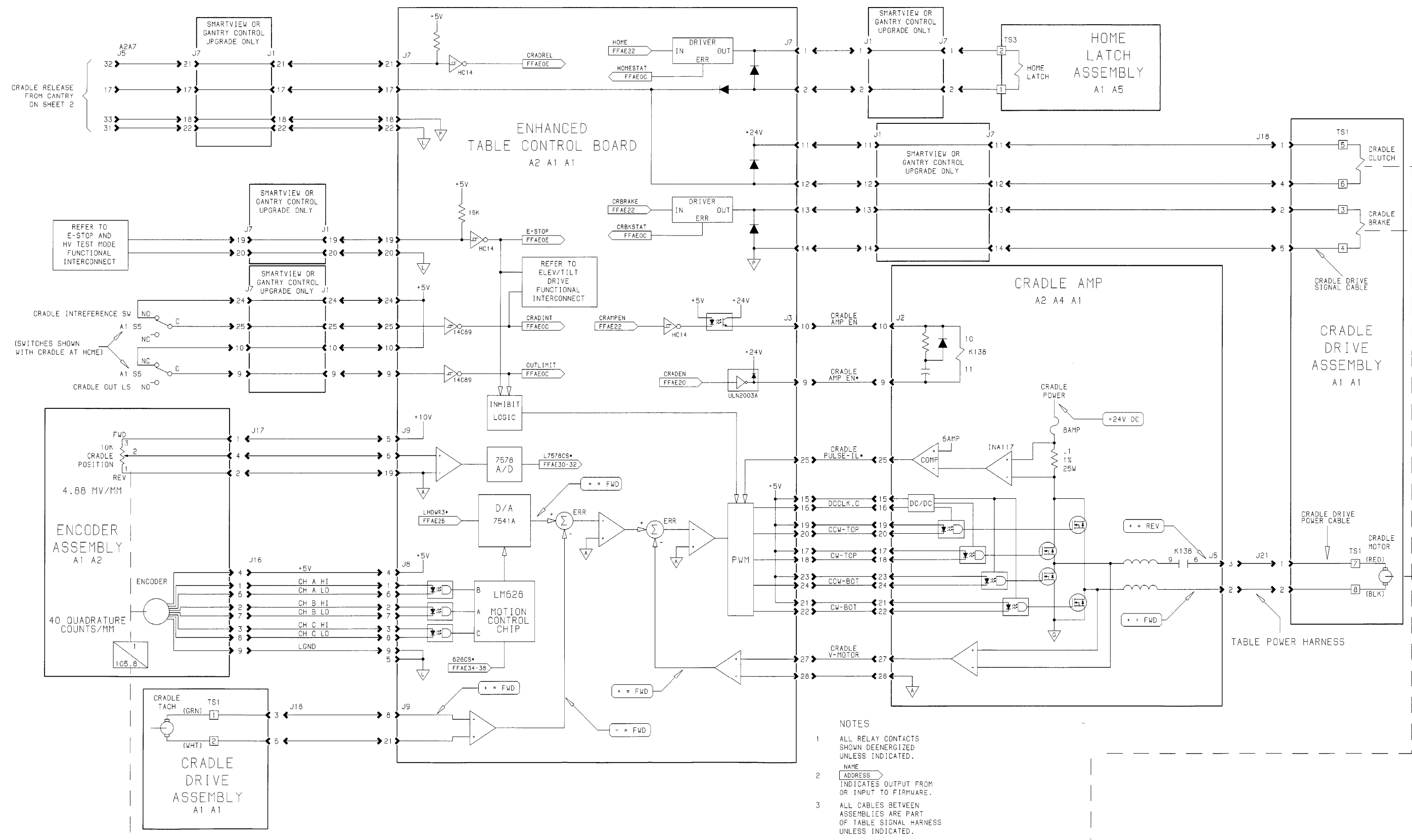


Figure 4-4 2267756 Sheet 1, Rev. 0

Section 5.0 –Table Cradle Drive (Vertical & Longitudinal Drives) - 2123540



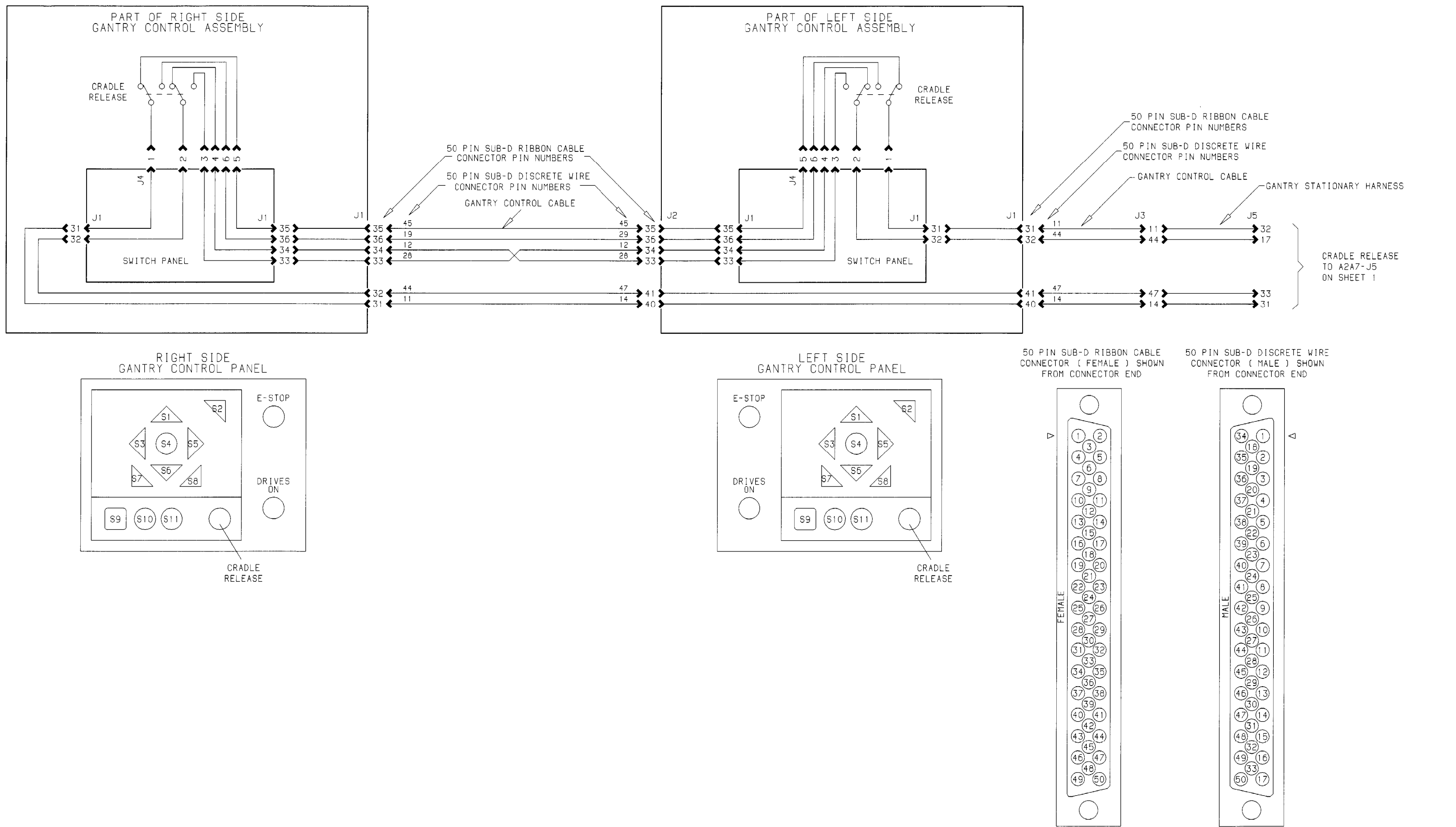
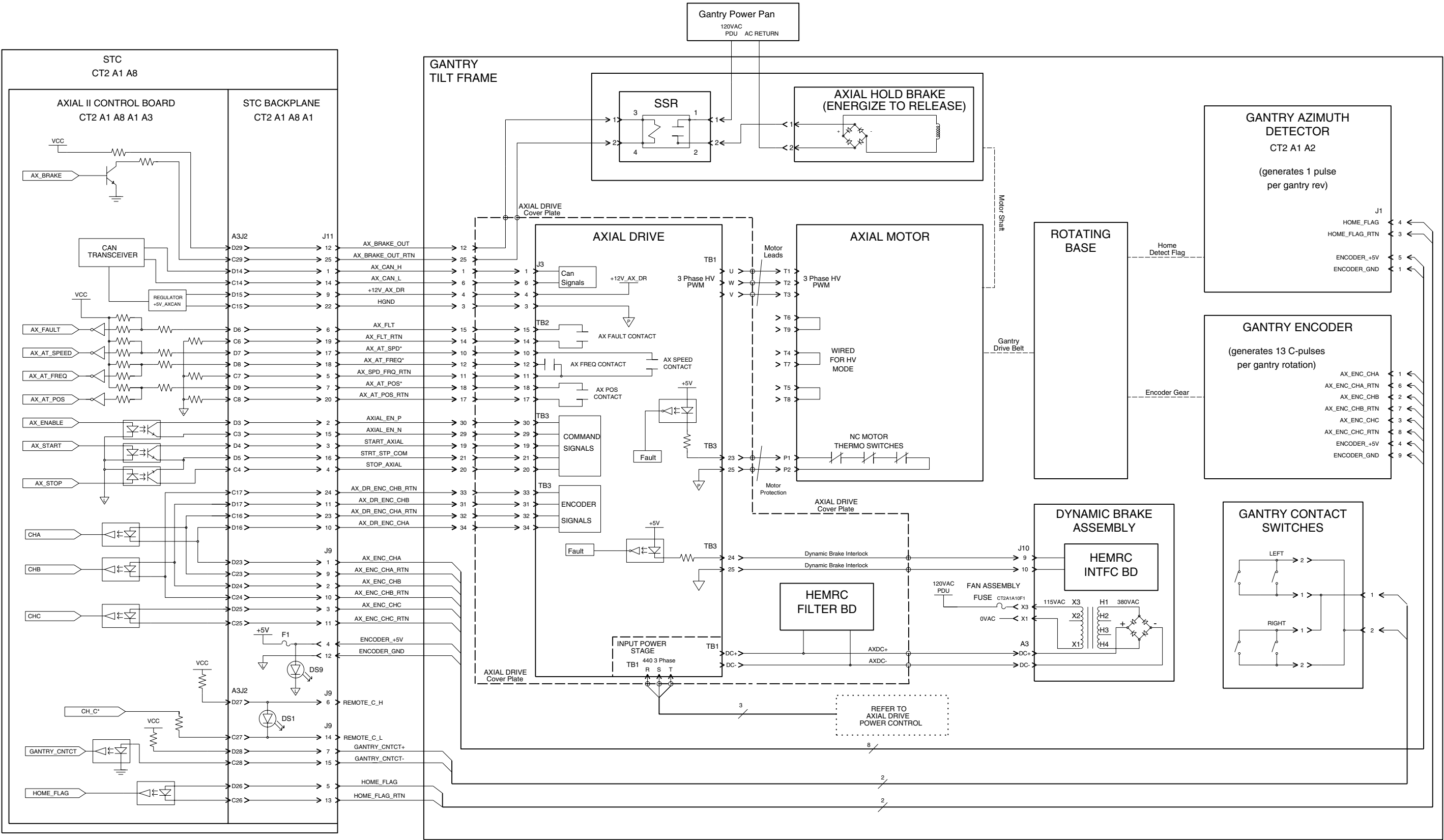


Figure 4-6 2123540 Sheet 2, Rev 1

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Chapter 5

Axial Control

Page 40 Section 1.0 –Axial Servo Control Loop - 2267757

Section 2.0 – Axial Drive Power Control - 2267784

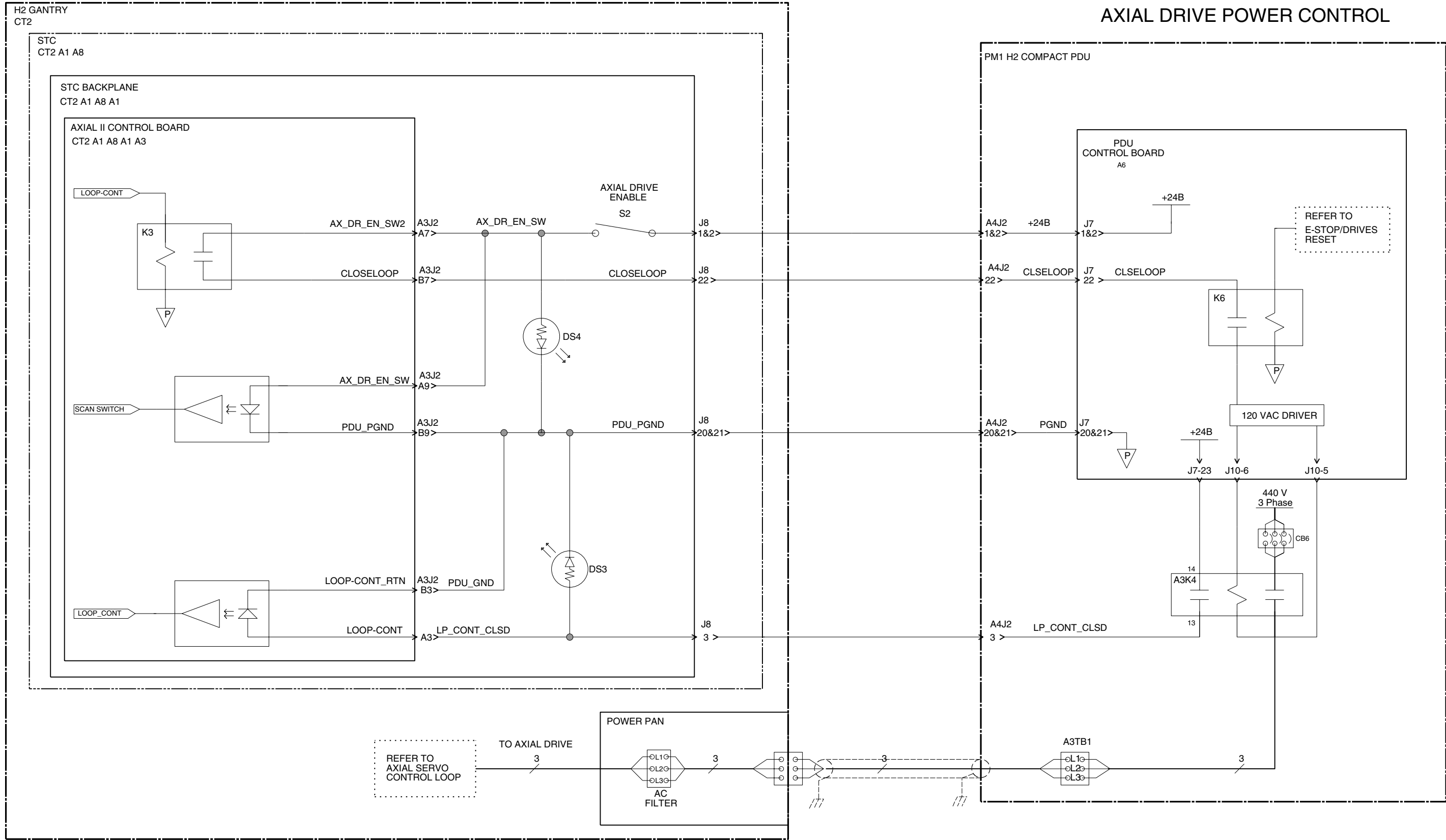


Figure 5-2 2267784 Sheet 1 Rev. 0

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Chapter 6

Operator I/O

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Page 44 Section 1.0 —Gantry Operator Interface - 2267760

Section 2.0 – Site X-Ray ON Light - 2267761

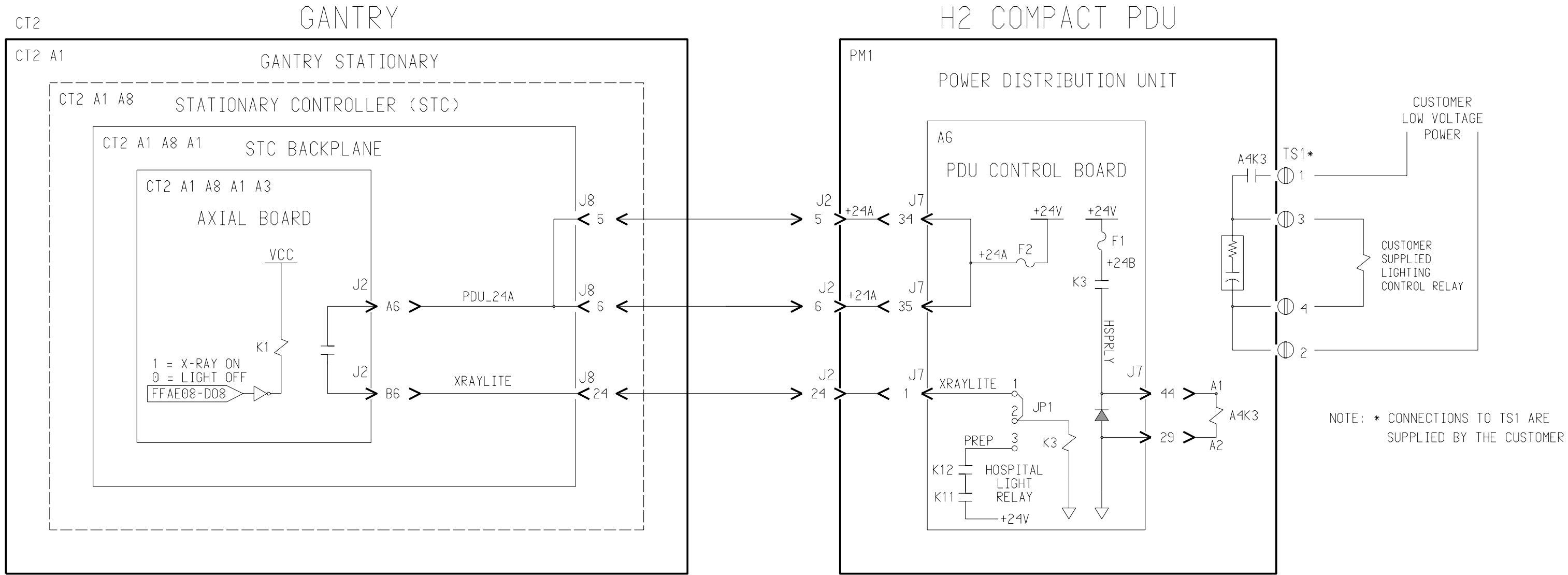
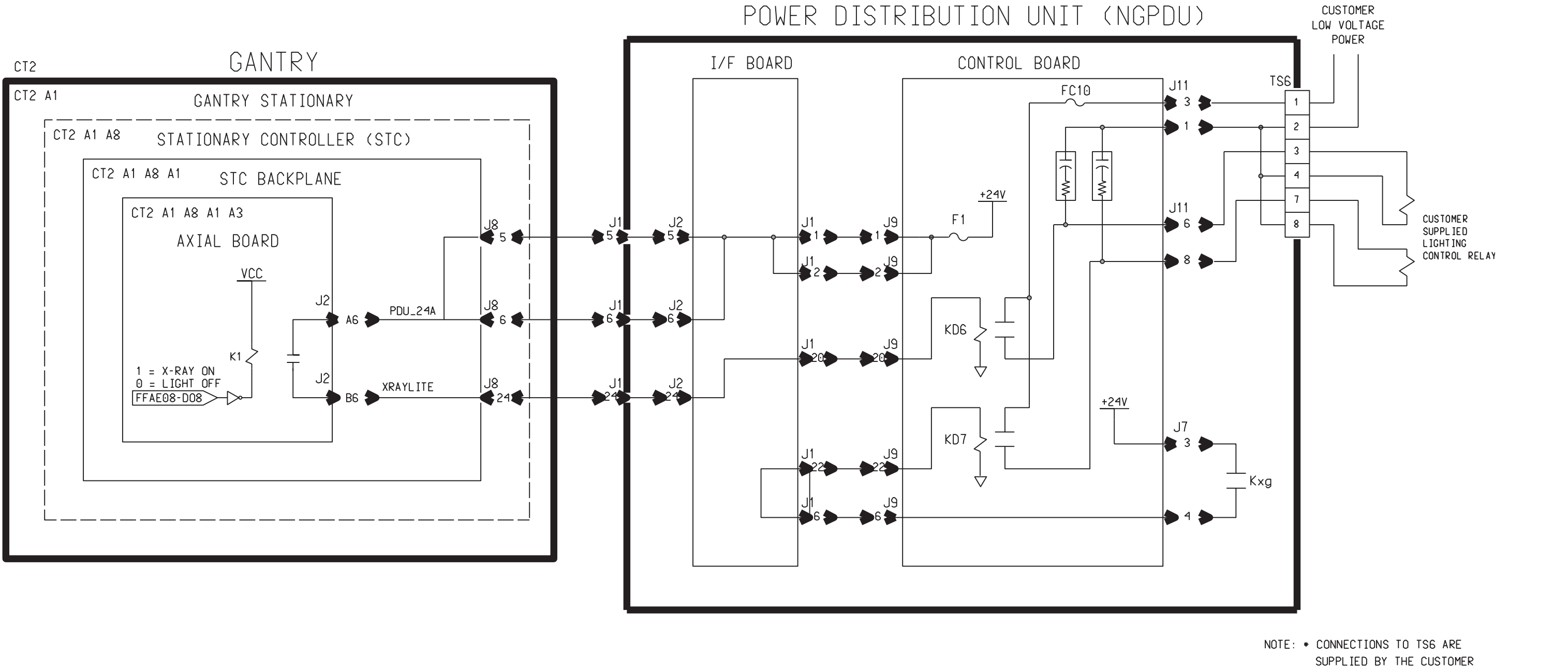


Figure 6-2 2267761 Sheet 1, Rev. 0 (CPDU shown; see Figure 6-3 for NGPDU)



NOTE: * CONNECTIONS TO TS6 ARE SUPPLIED BY THE CUSTOMER

Figure 6-3 Site X-Ray ON Light (NGPDU shown; see [Figure 6-2](#) for CPDU)

Section 3.0 – Site System ON Light

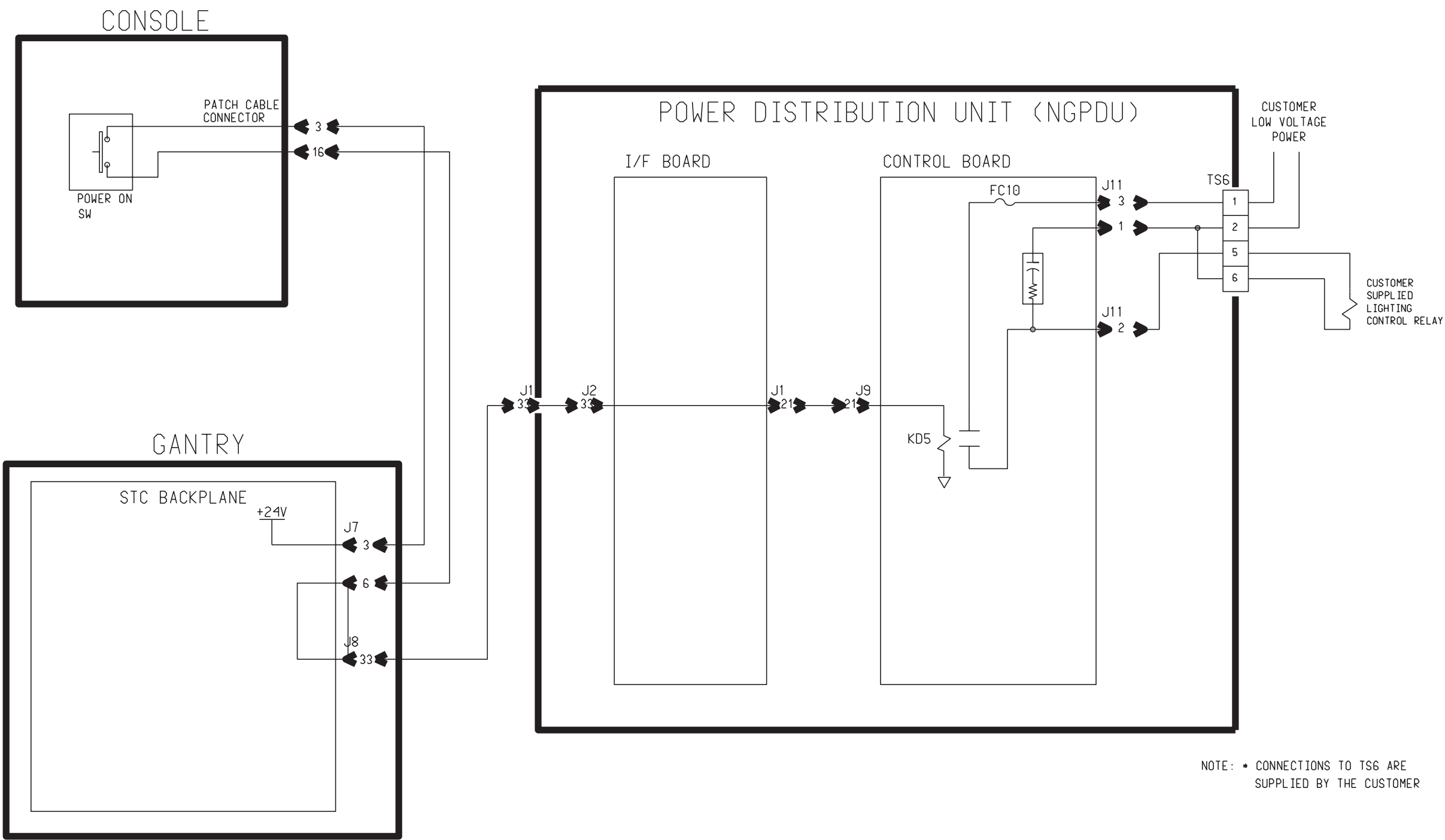


Figure 6-4 Site System ON Light (NGPDU Shown)

Section 4.0 — RX Tilt & Console Gantry Reset - 2267765

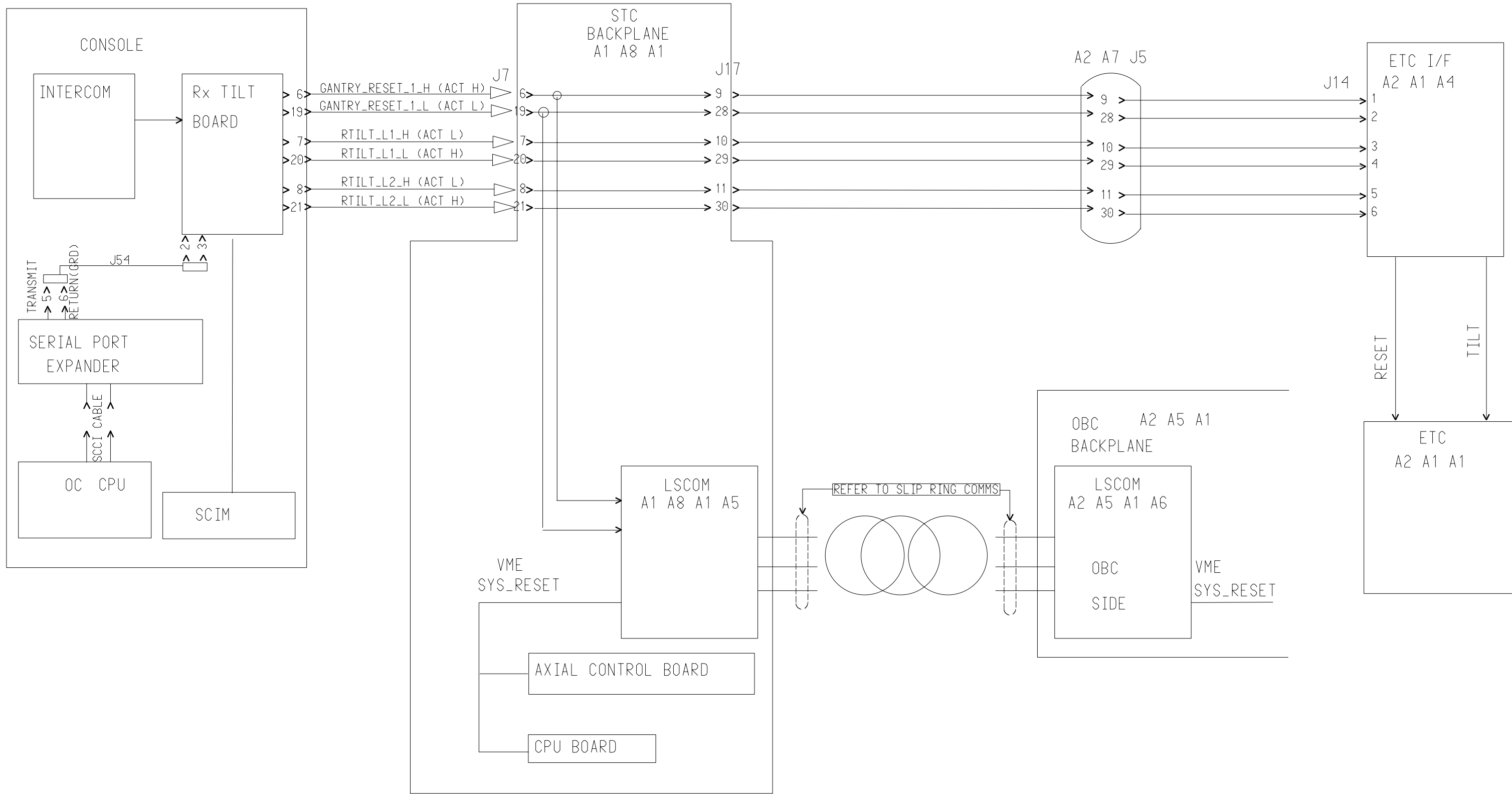


Figure 6-5 2267765 Sheet 1, Rev. 0

Chapter 7

System Monitoring

Section 1.0 — Mains Undervoltage - 2267731 (Not Applicable to NGPDU)

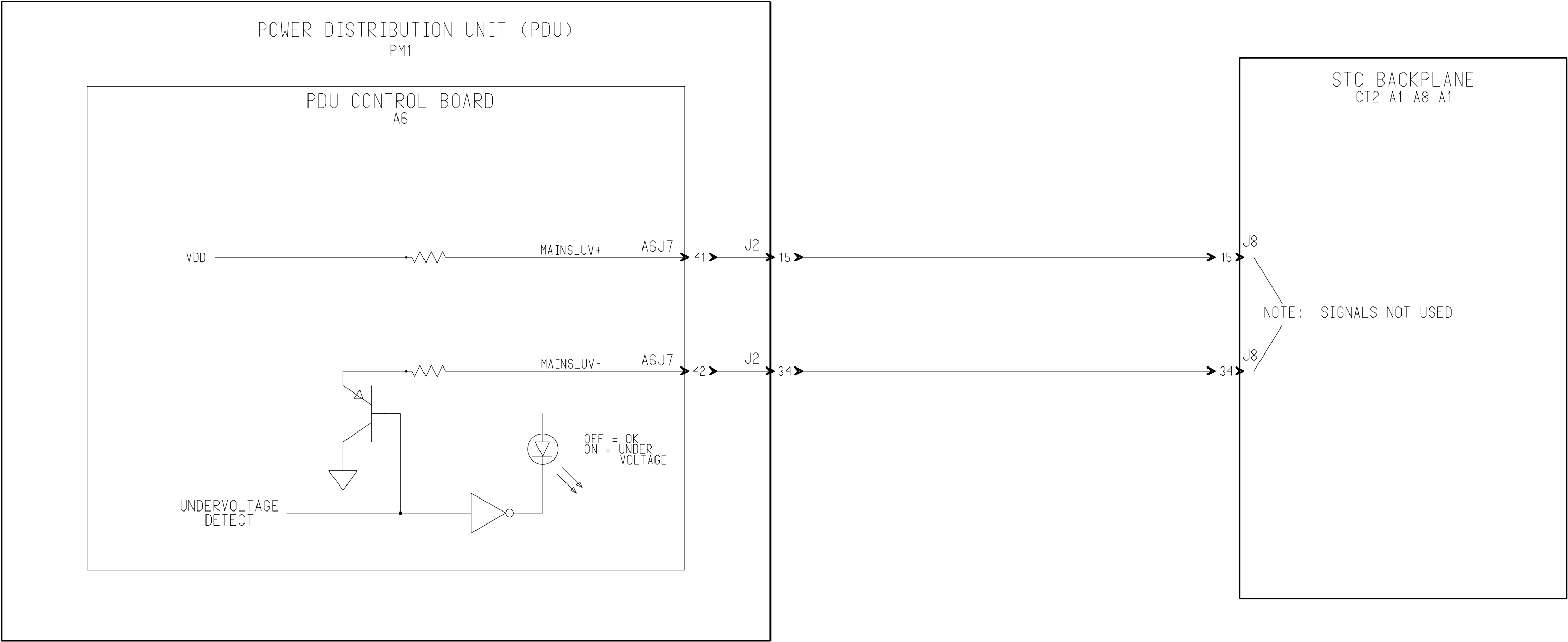
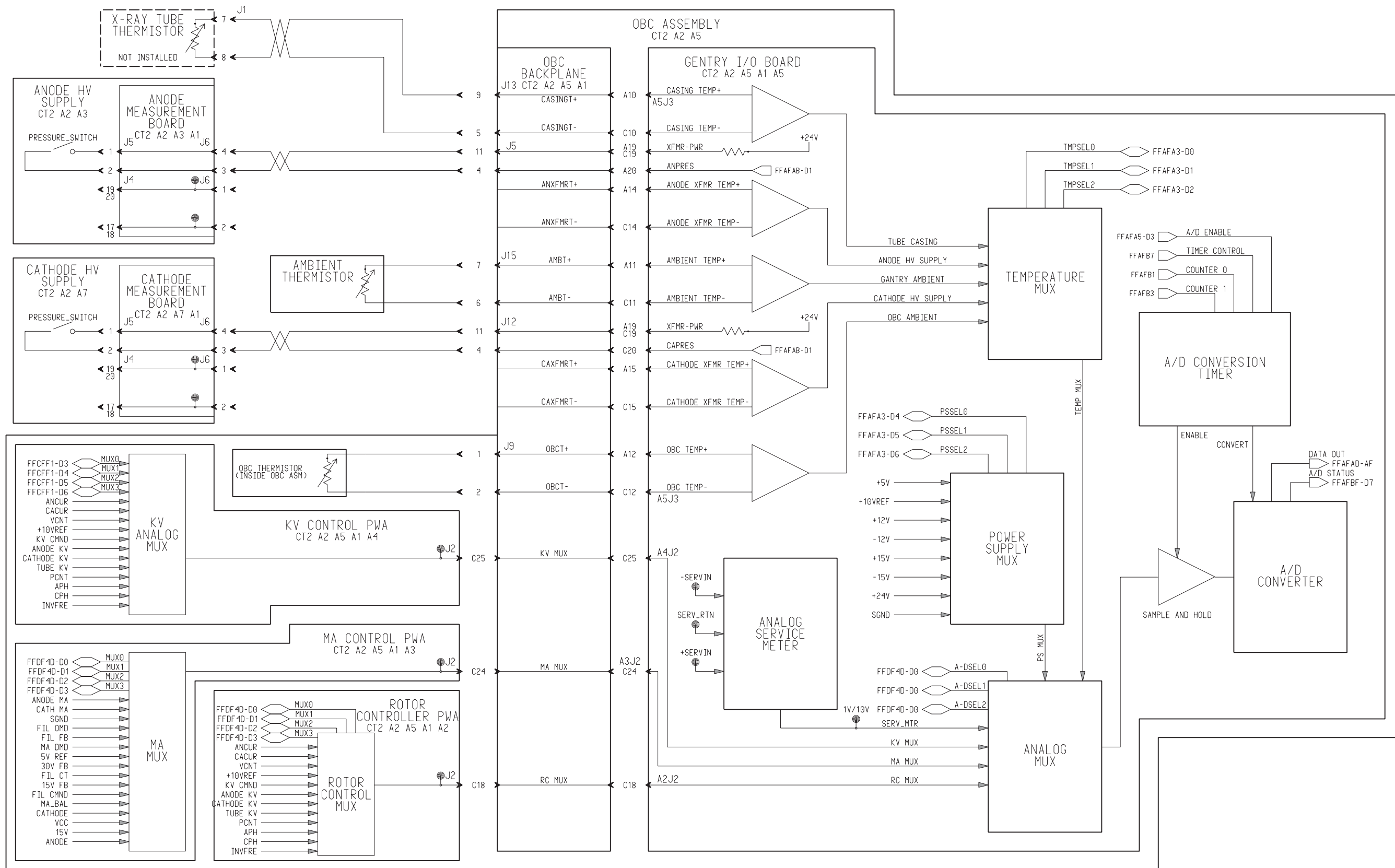


Figure 7-1 2267731 Sheet 1, Rev. 0

Figure 7-2 2267766 Sheet 1, Rev. 0



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Chapter 8

System Power Control

Section 1.0 – HVDC Bus Control - 2267769

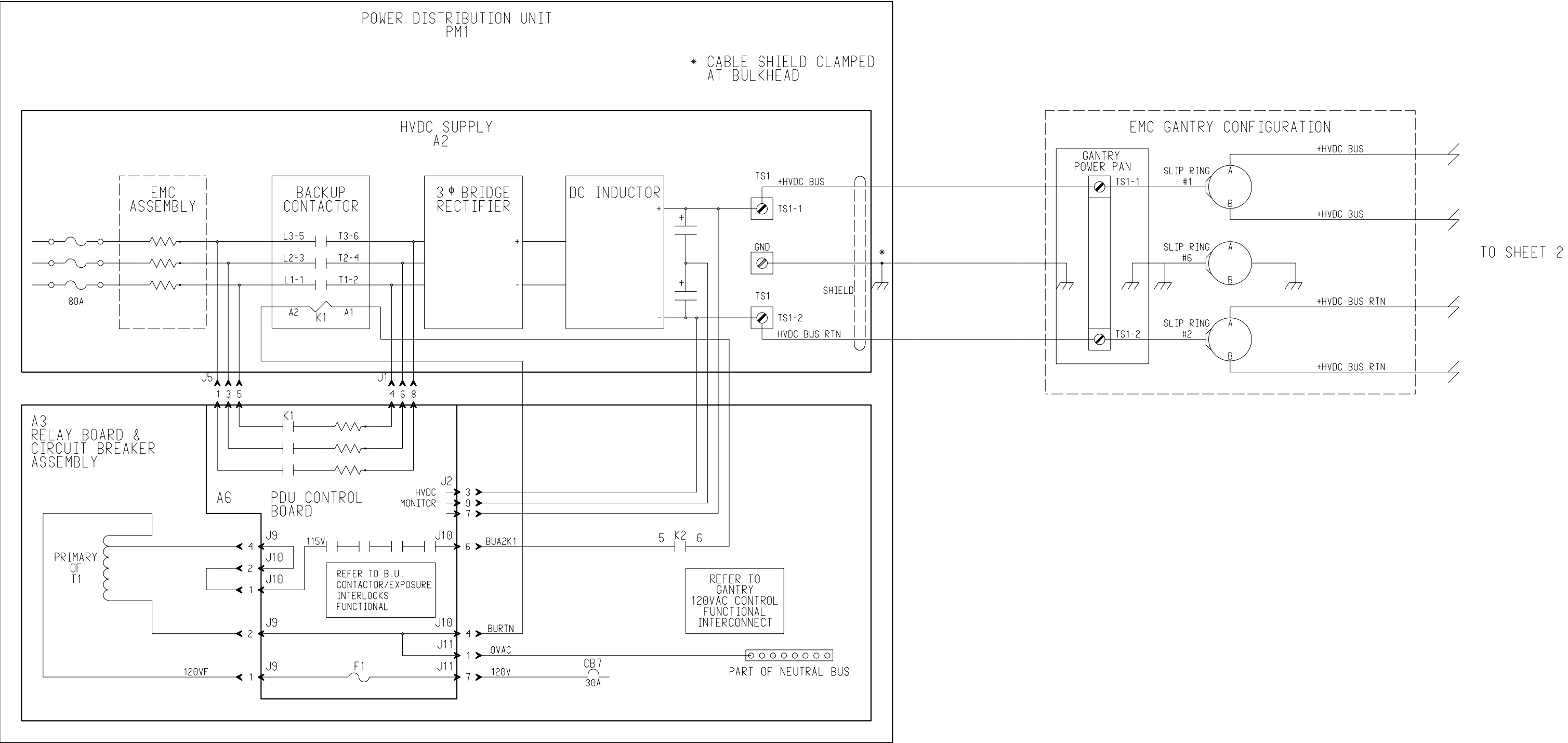


Figure 8-1 2267769 Sheet 1, Rev. 0 (CPDU shown; see [Figure 8-2](#) for NGPDU)

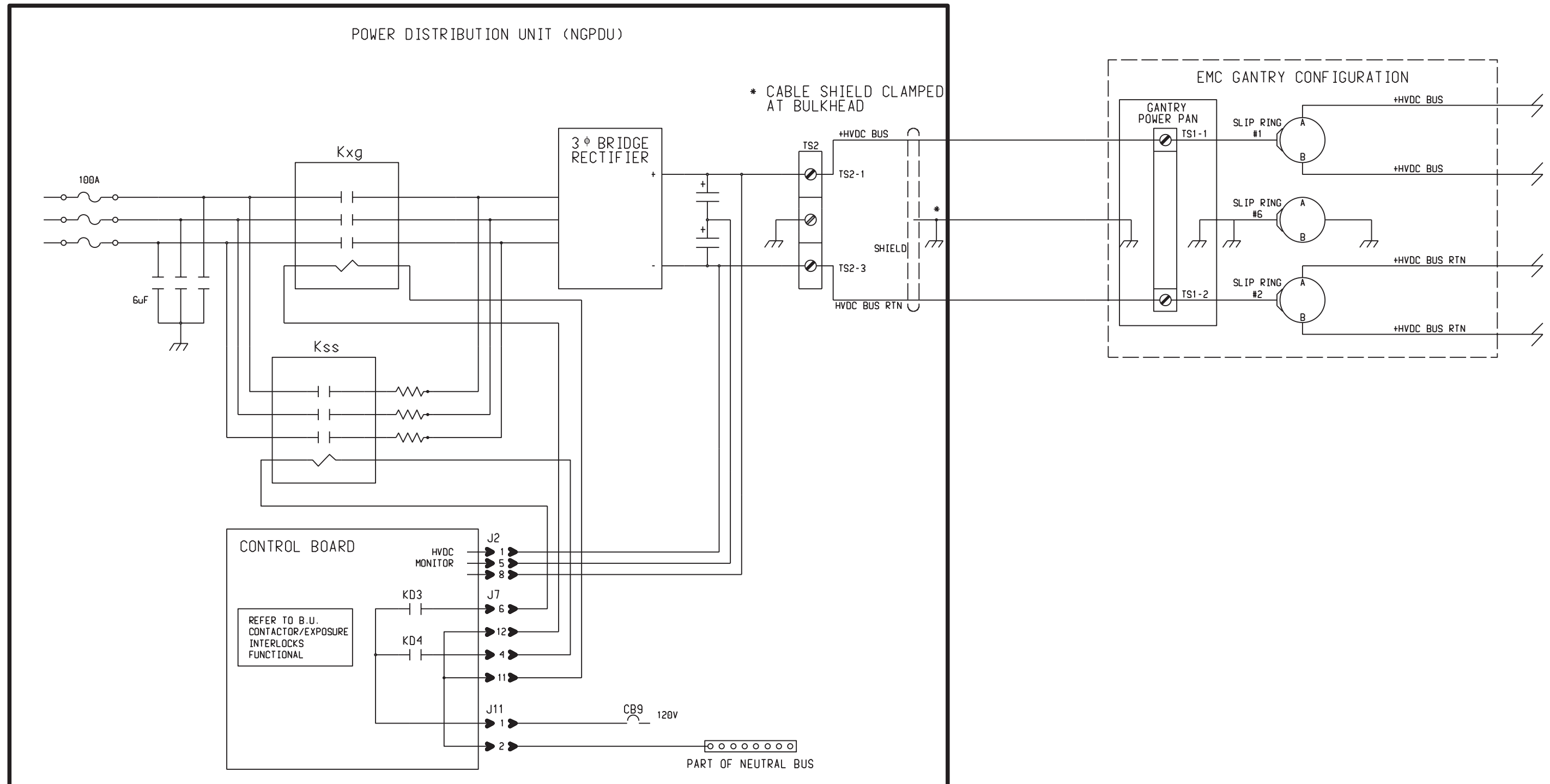


Figure 8-2 HVDC Bus Control (NGPDU shown; see Figure 8-1 for CPDU)

SYSTEMS WITH PERFORMIX TUBE

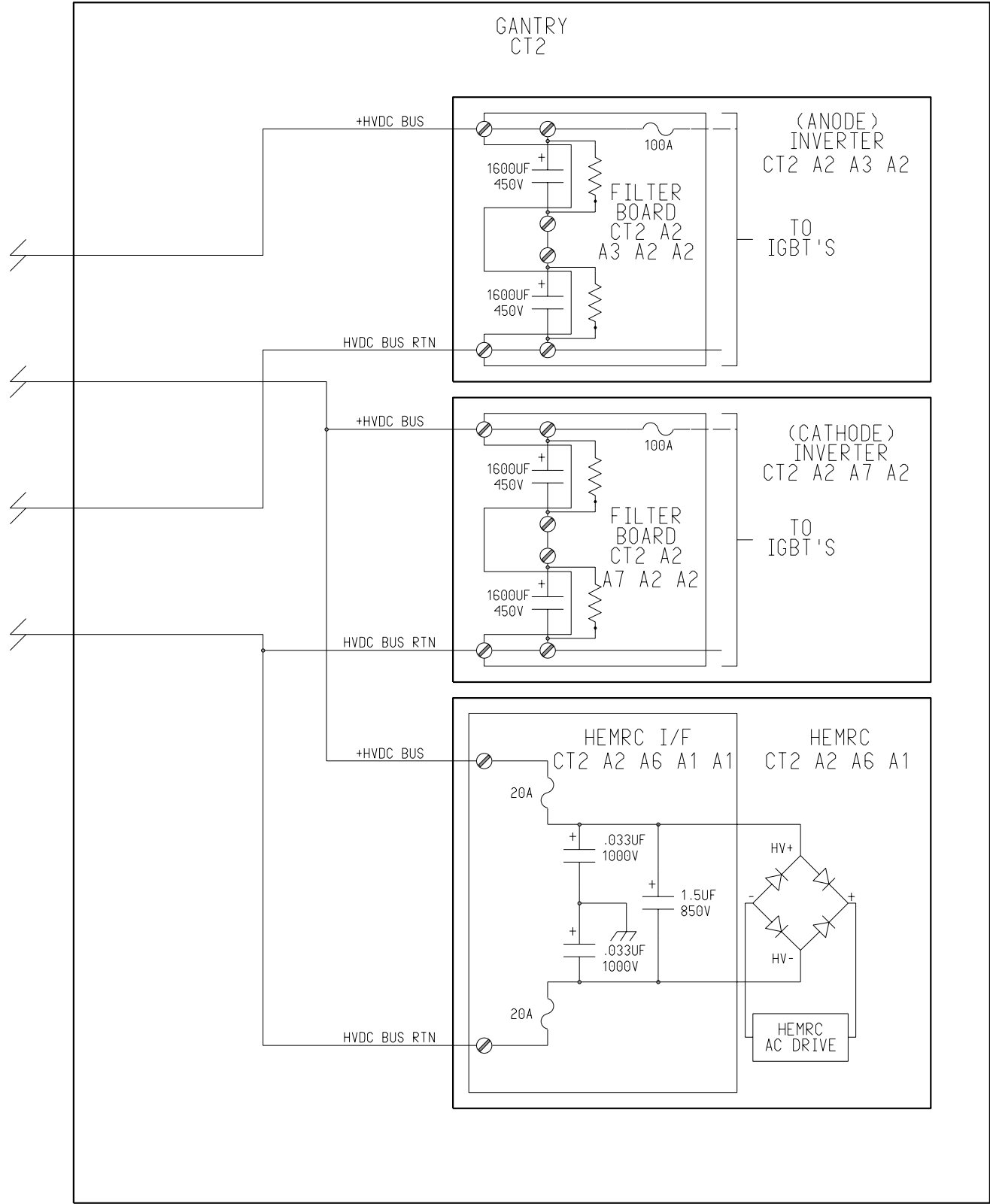


Figure 8-3 2267769 Sheet 2, Rev. 0

Section 2.0 – Gantry 120VAC Control - 2267770

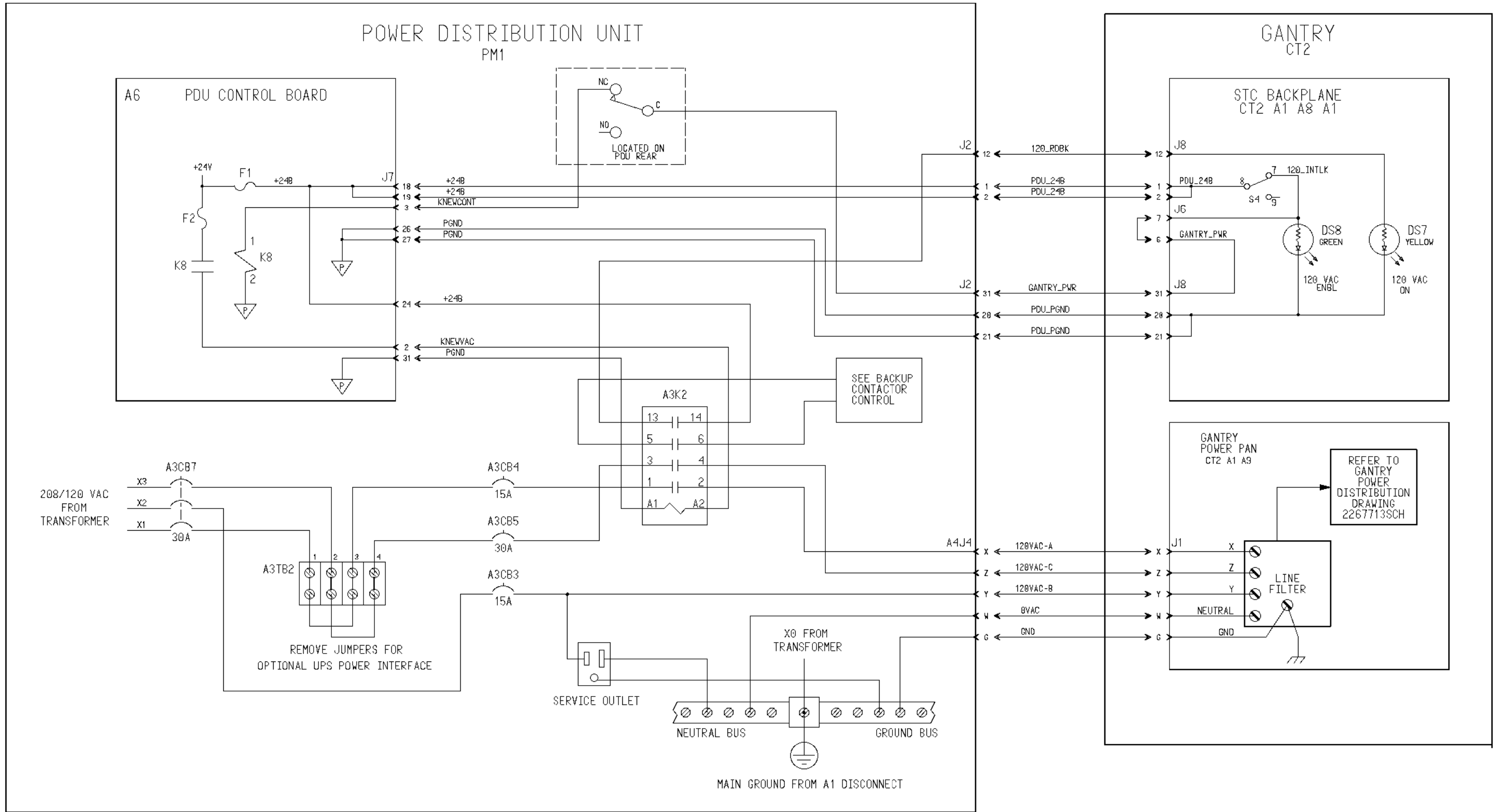


Figure 8-4 2267770 Sheet 1, Rev. 1 (CDPU shown; see Figure 8-5 for NGPDU)

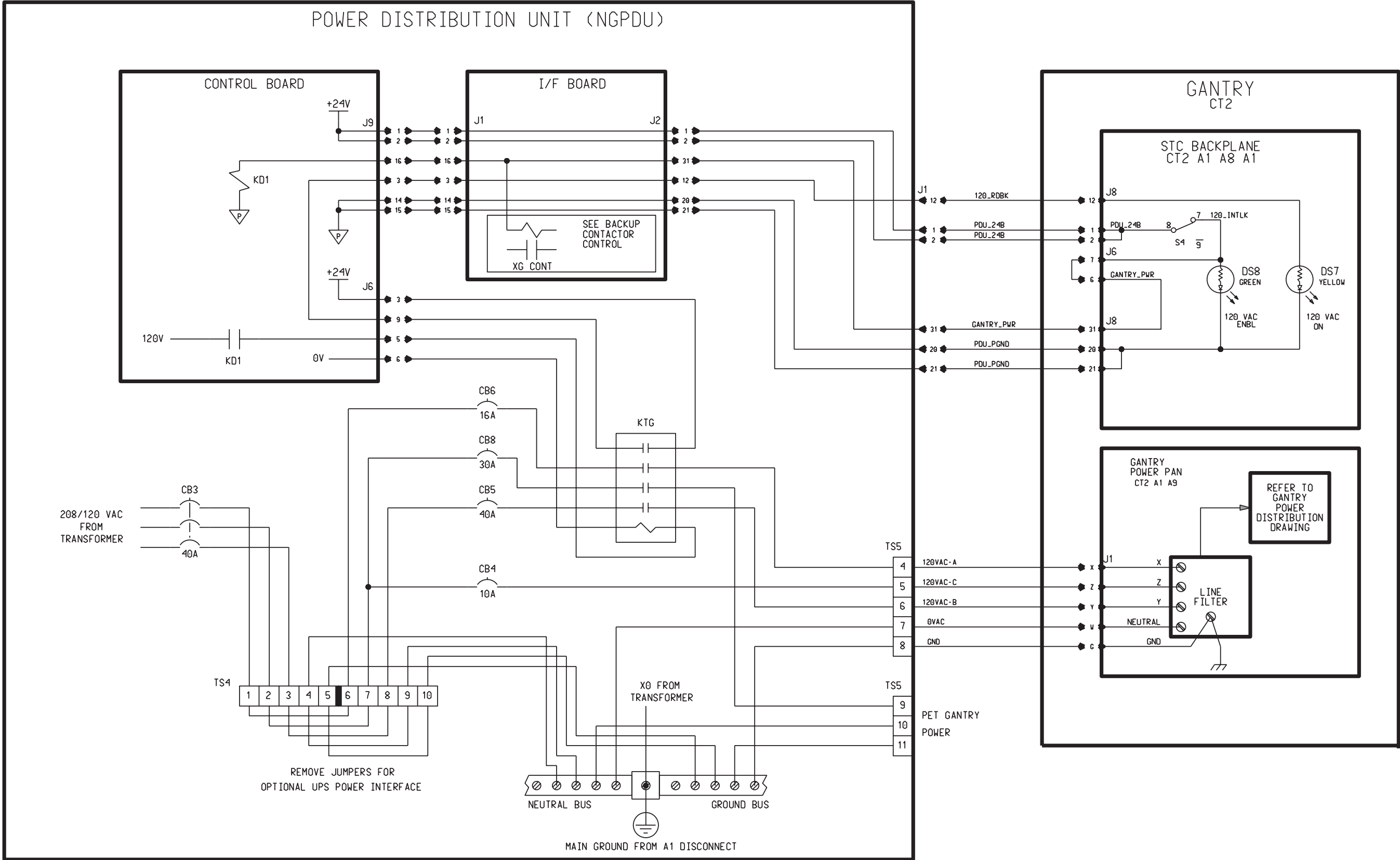
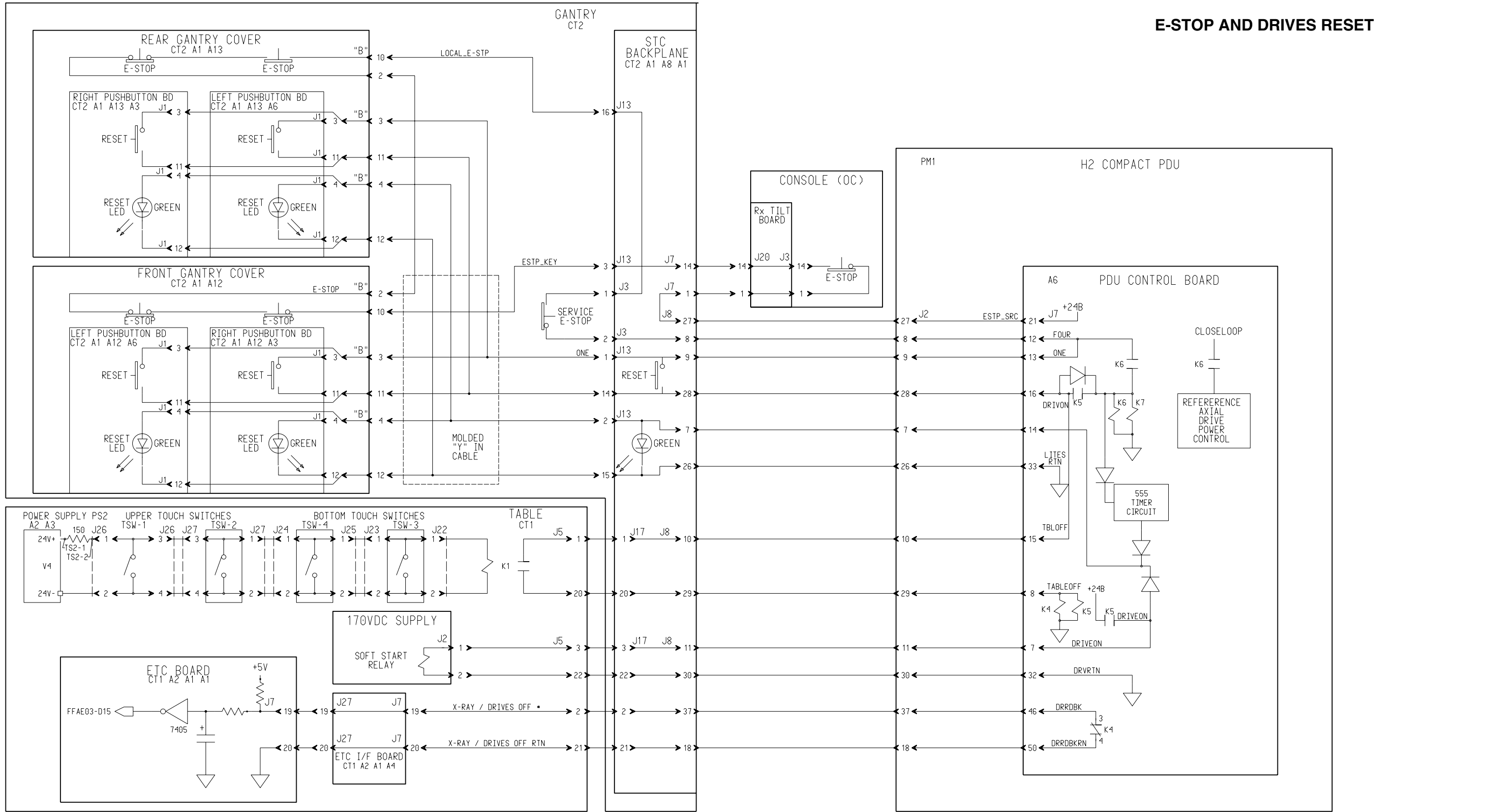


Figure 8-5 Gantry 120VAC Control (NGPDU shown; see [Figure 8-4](#) for CPDU)

Section 3.0 – E-Stop and Drives Reset - 2267773



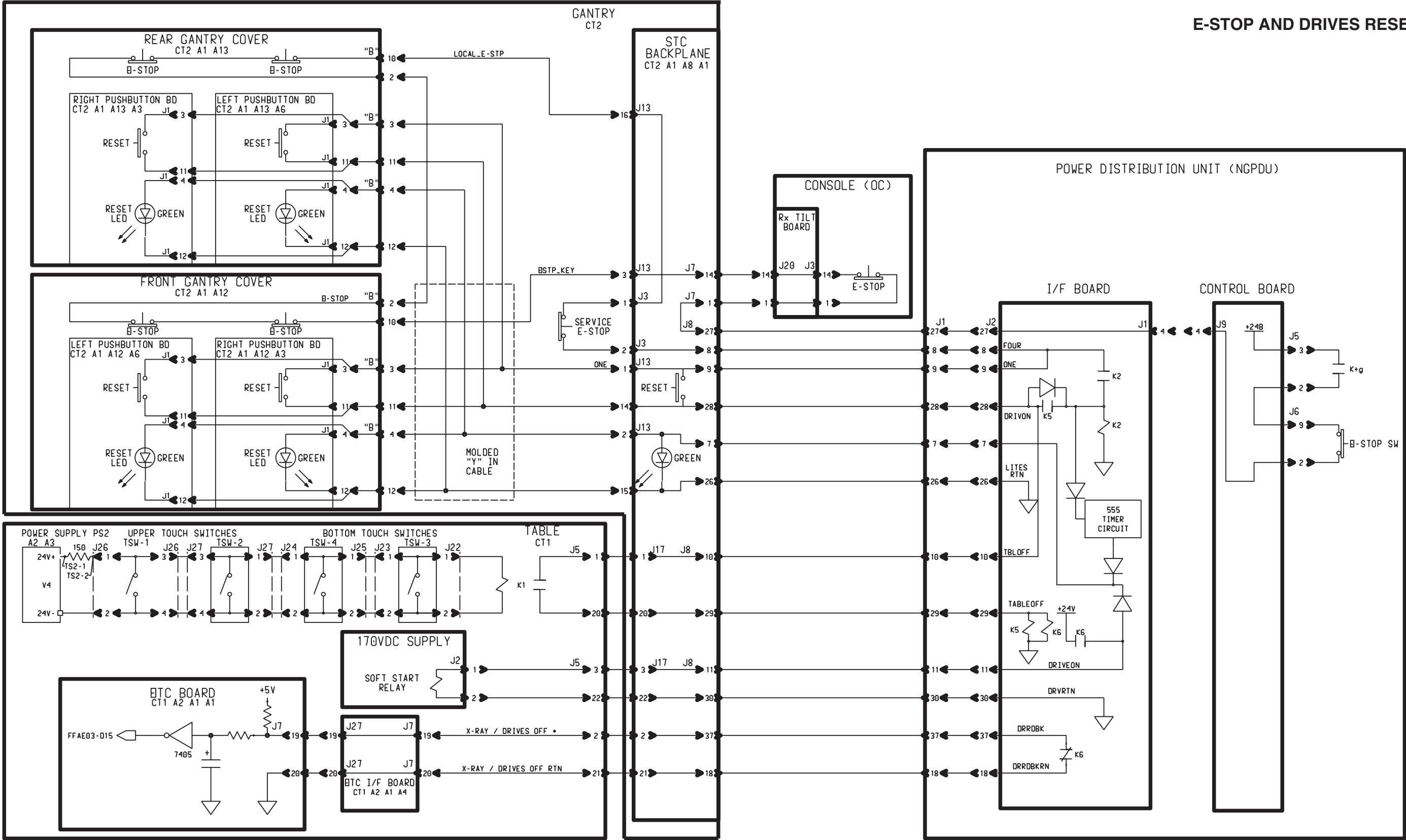


Figure 8-7 E-Stop and Drives Reset (NGPDU shown; see [Figure 8-6](#) for CPDU)

Section 4.0 – Gantry Stationary Power and Ground - 2267713

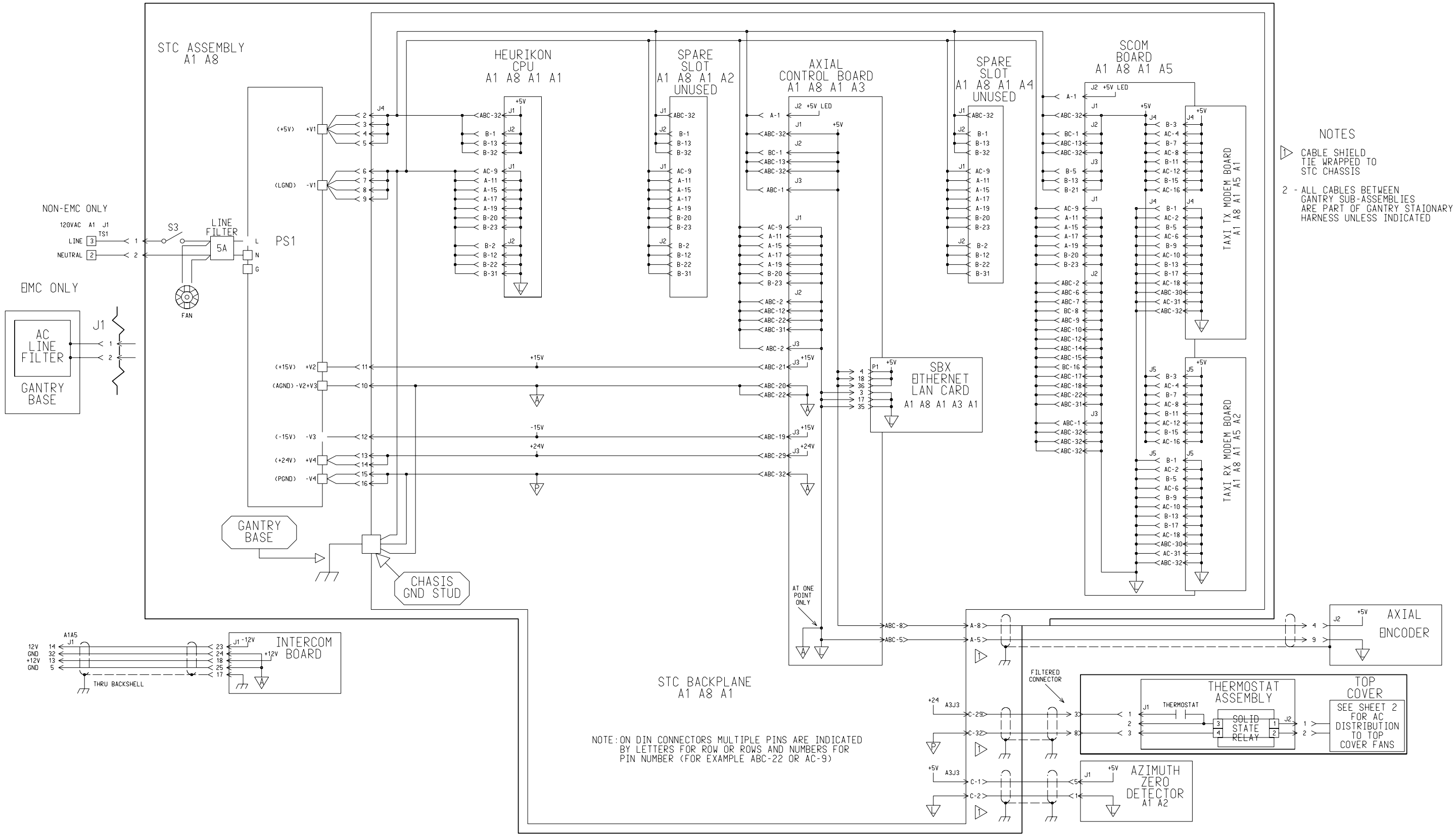
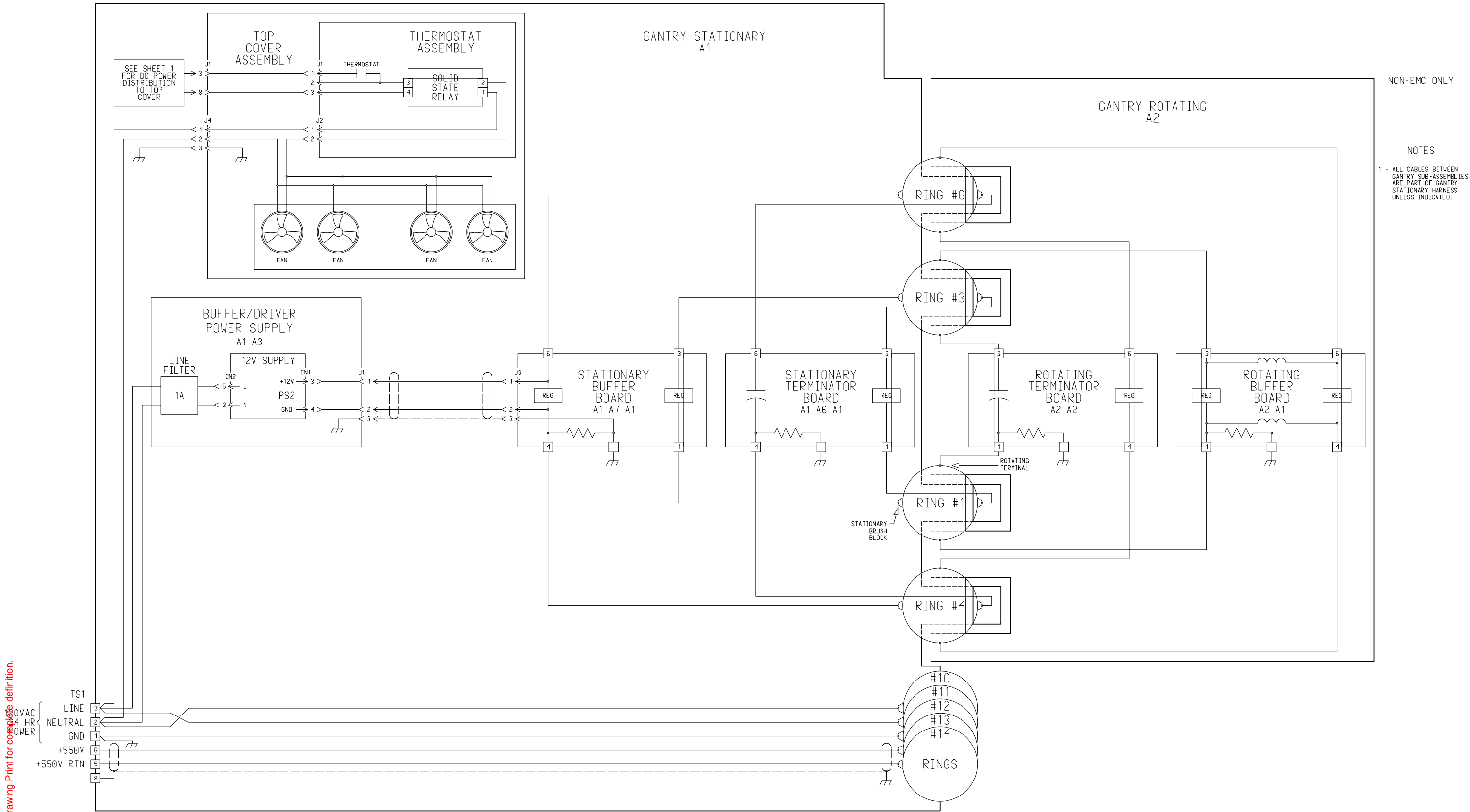


Figure 8-8 2267713 Sheet 1, Rev. 1



rawing Print for complete definition.

Figure 8-9 2267713 Sheet 2, Rev. 1

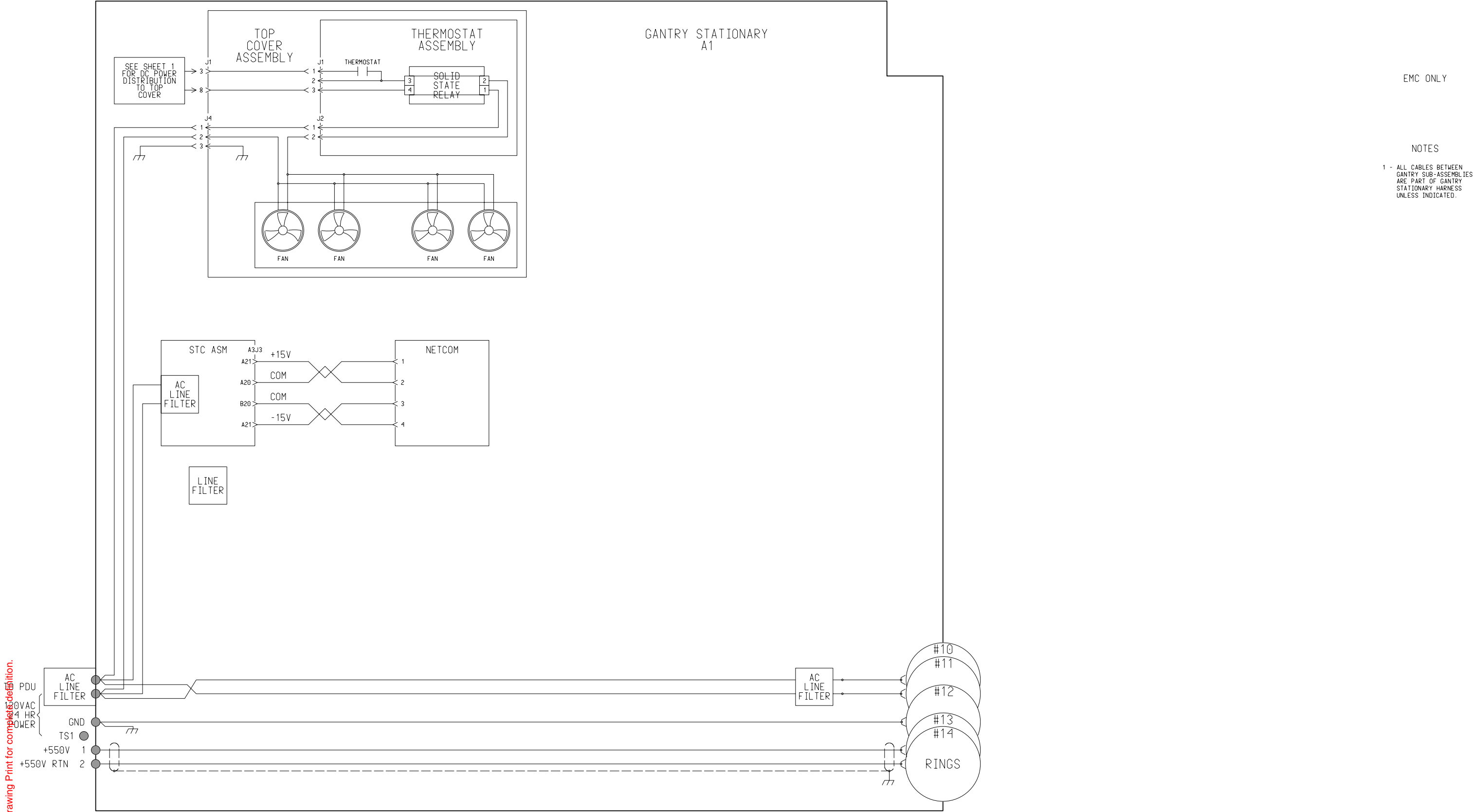


Figure 8-10 2267713 Sheet 3, Rev. 1

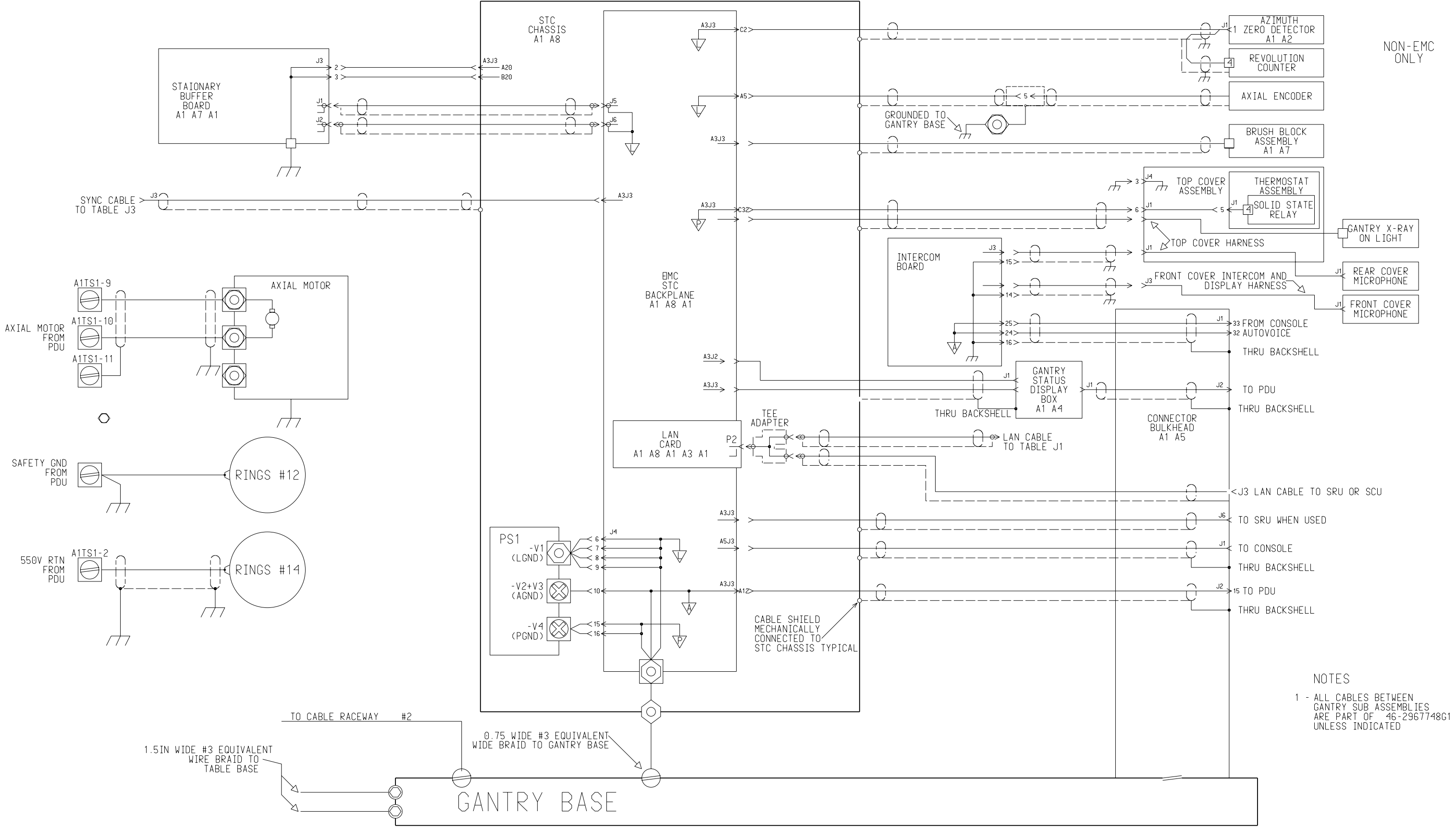


Figure 8-11 2267713 Sheet 4, Rev. 1

Chapter 9

Image Generation (Global Console - Linux)

Section 1.0 – Scan Data Save

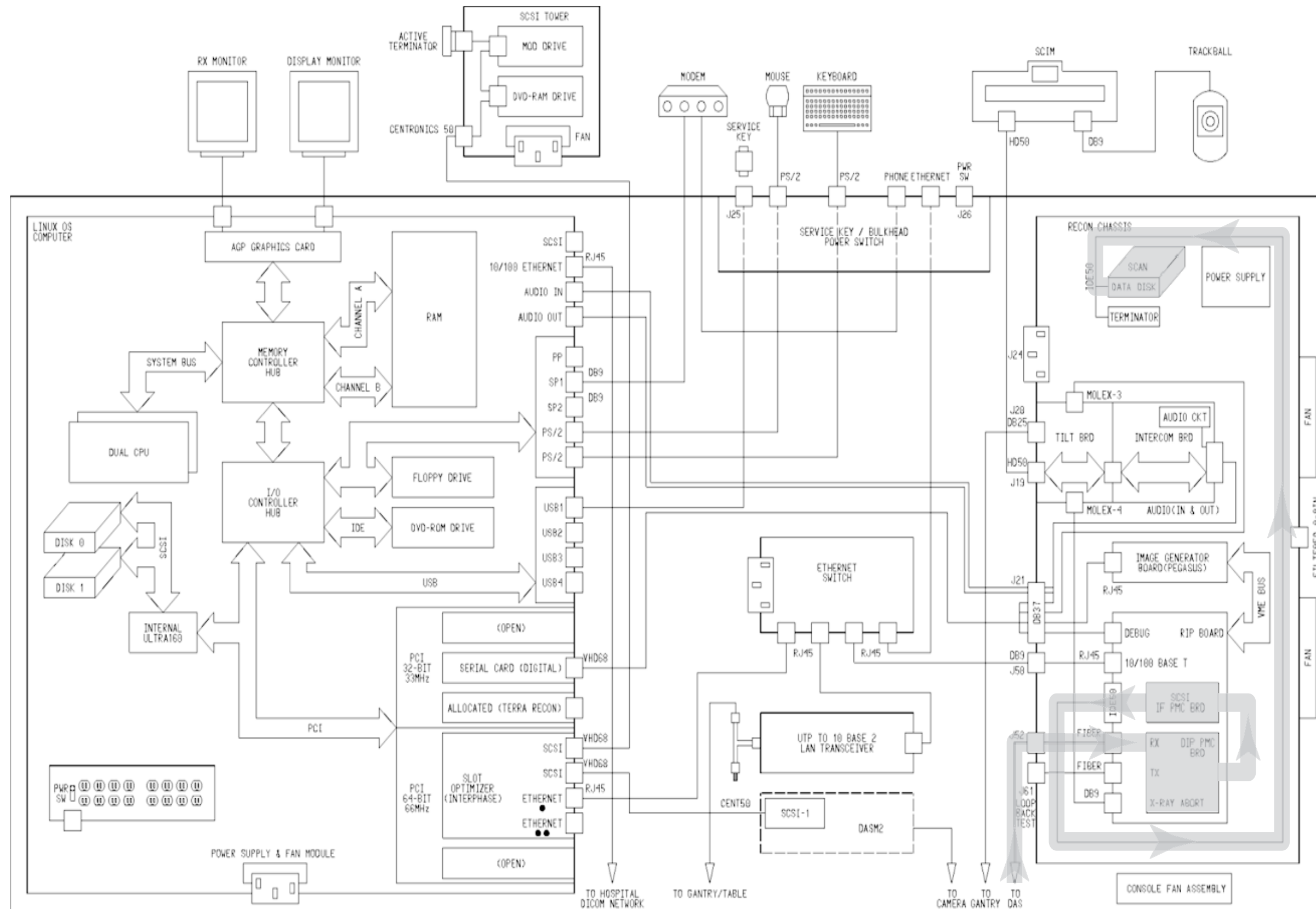


Figure 9-1 Scan Data Save

Section 2.0 – Data Restore

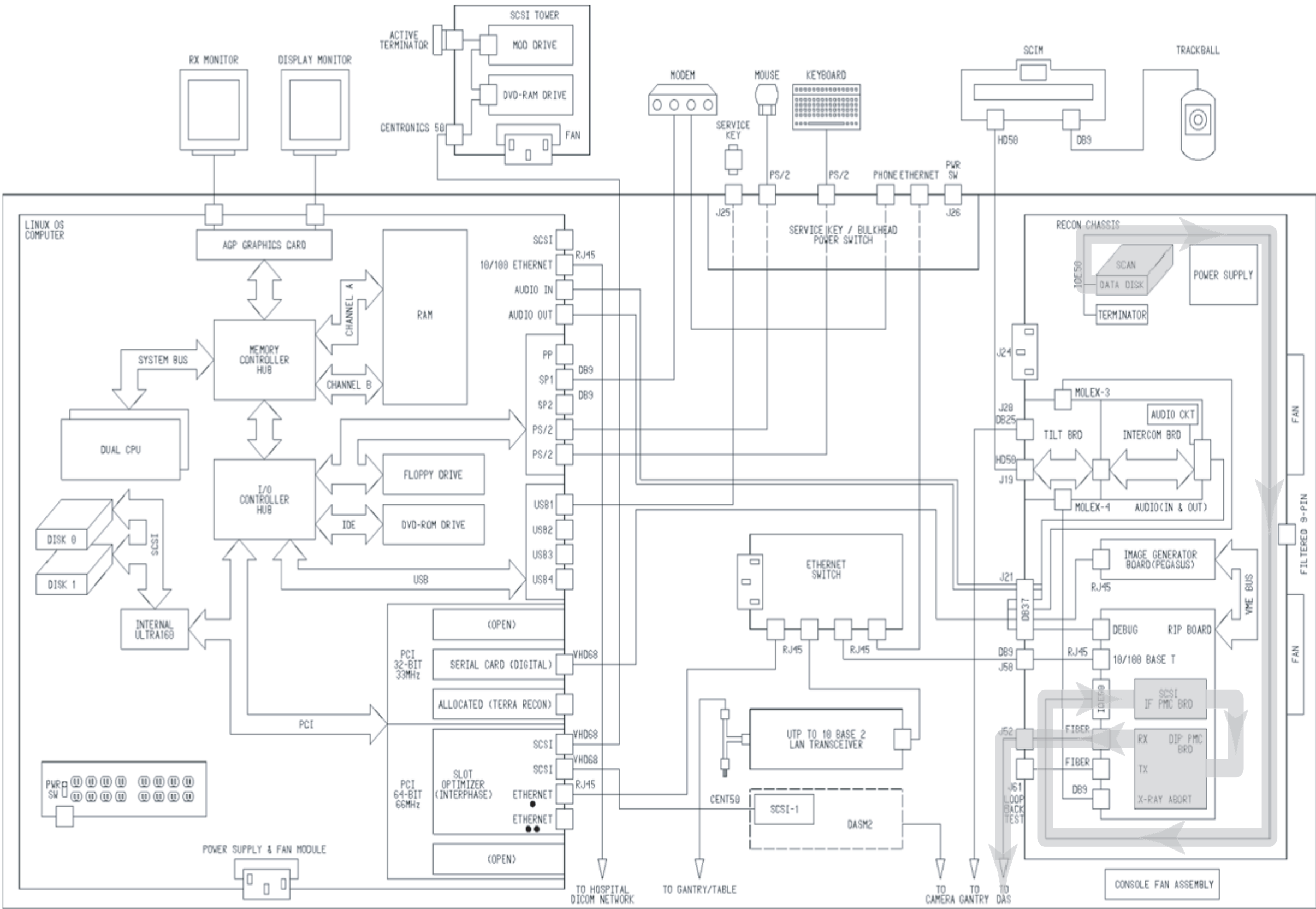


Figure 9-2 Data Restore

Section 3.0 — Pre-Processing, Filtered Back Projection & Post Processing

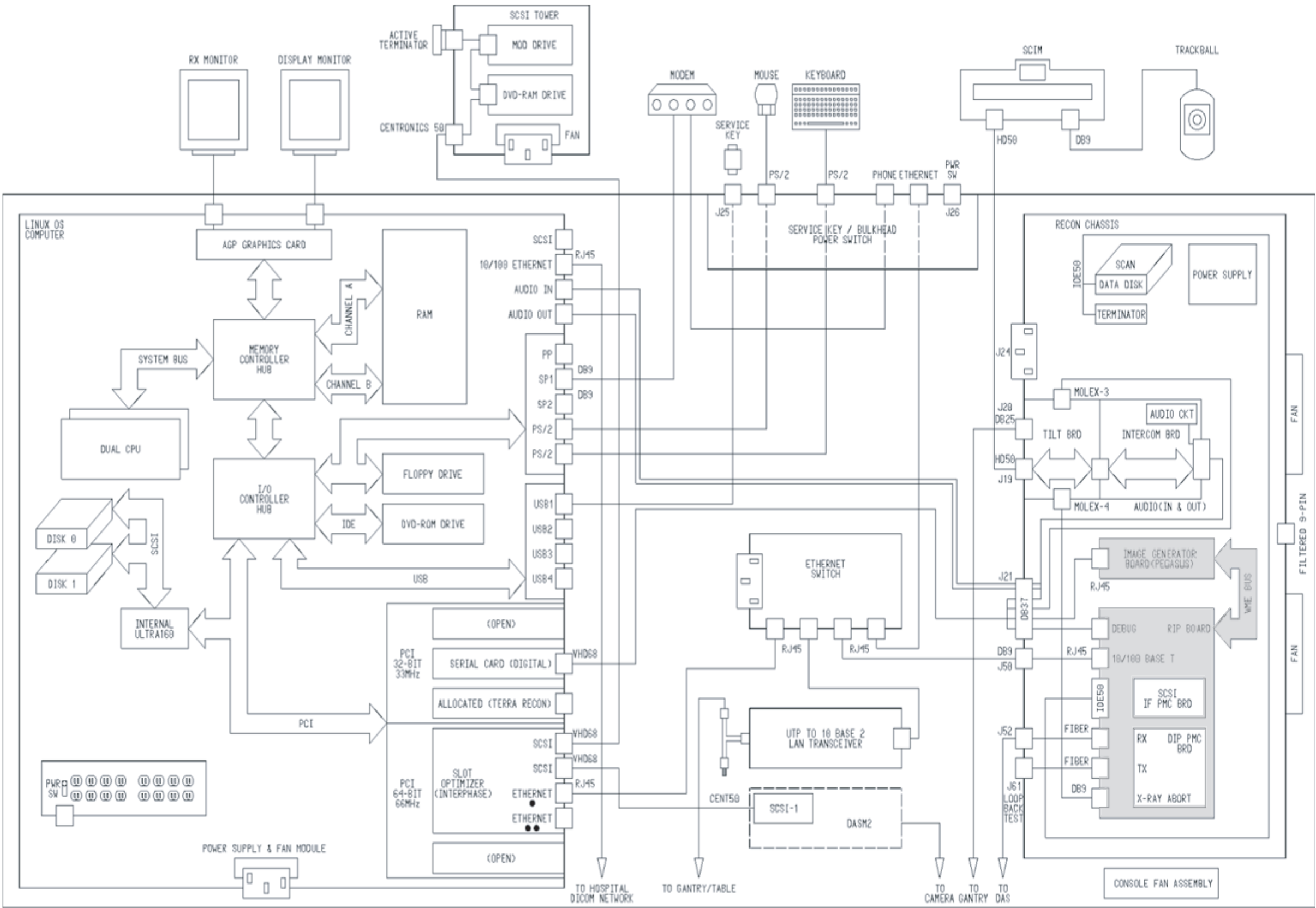


Figure 9-3 Pre-Processing, Filtered Back Projection & Post Processing

Section 4.0 – Image Transfer

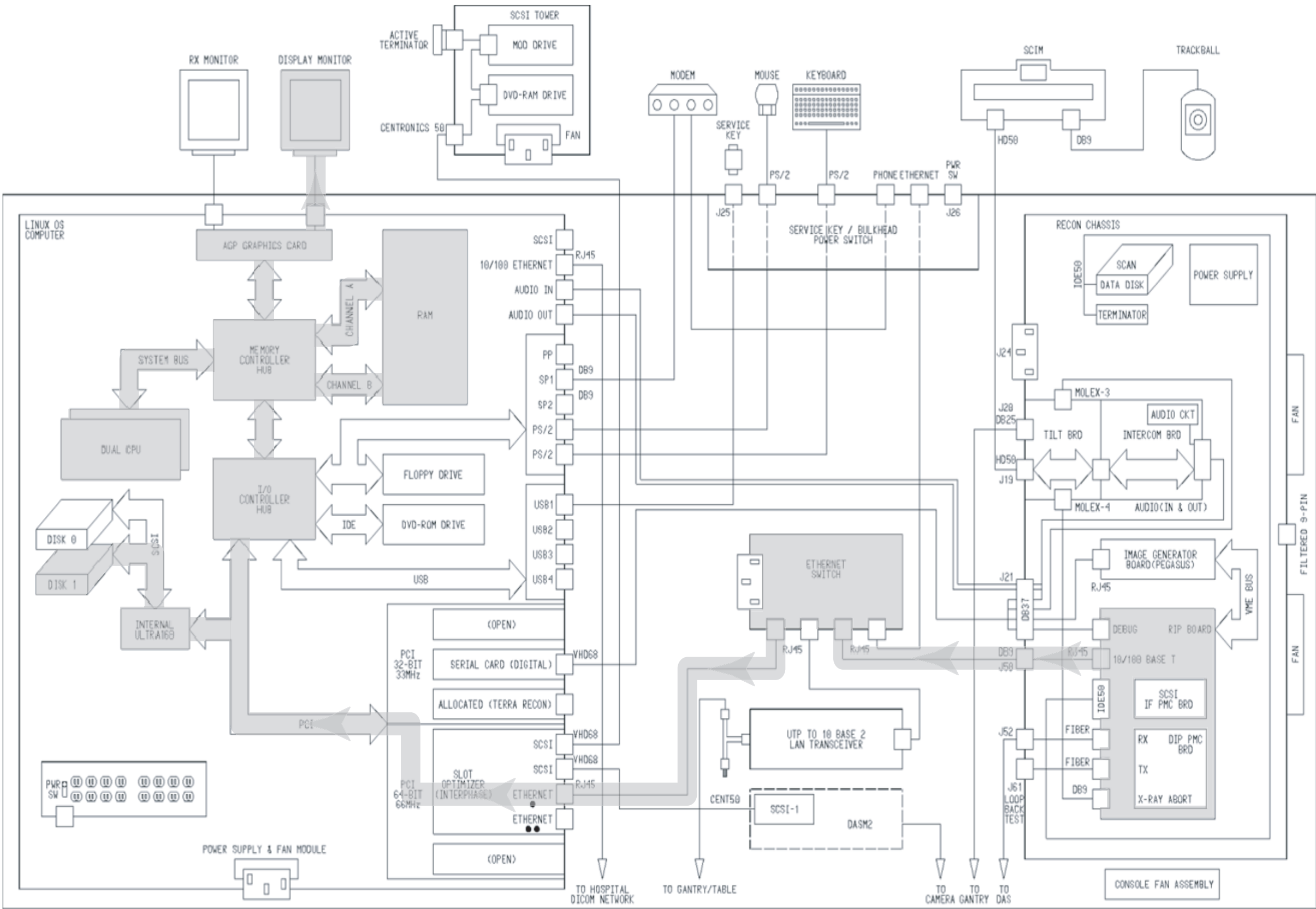


Figure 9-4 Image Transfer

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Chapter 10

Image Generation (Global Console - Octane2)

Section 1.0 – Scan Data Save

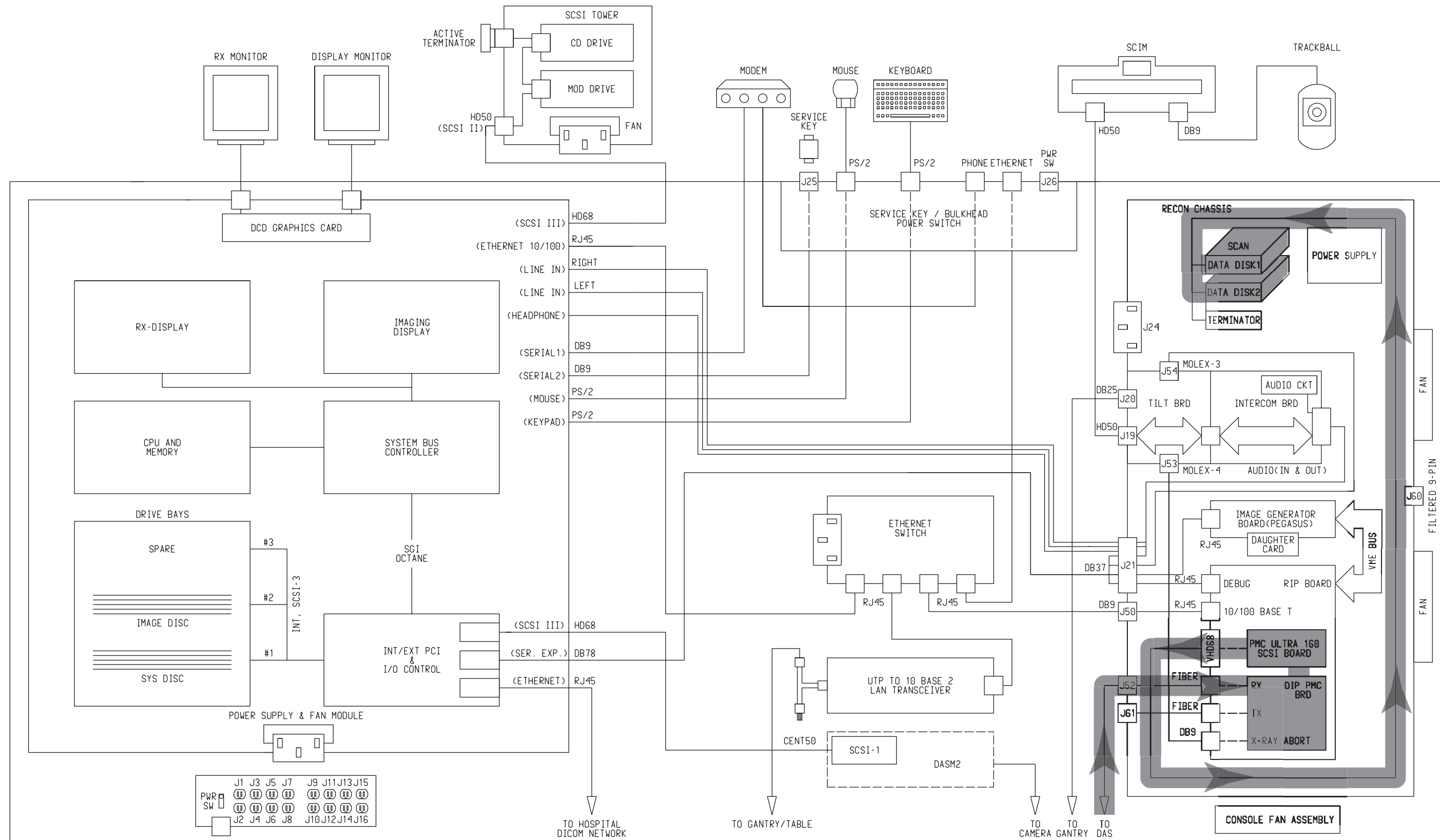


Figure 10-1 Scan Data Save

Section 2.0 – Data Restore

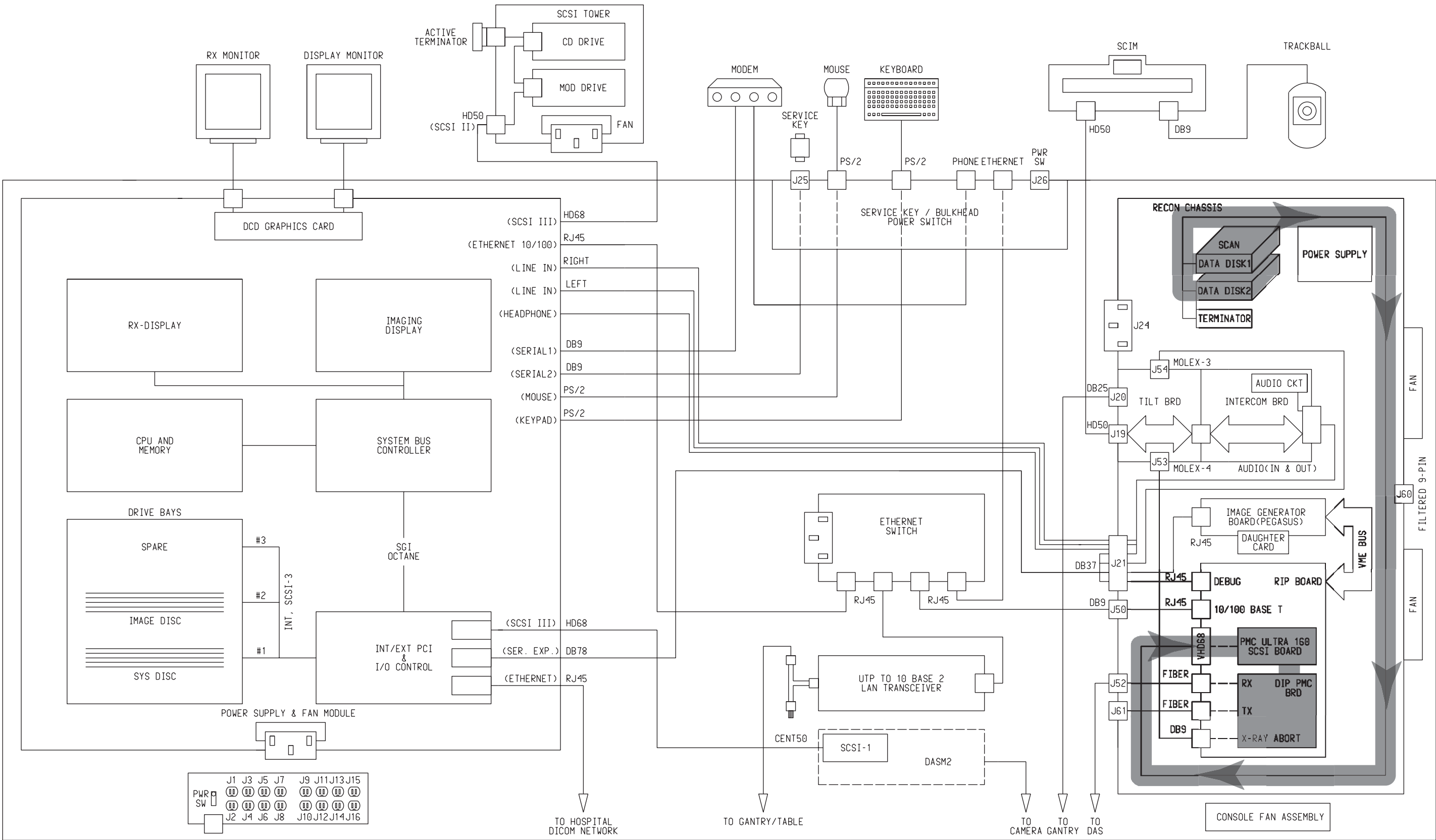


Figure 10-2 Data Restore

Section 3.0 — Pre-Processing, Filtered Back Projection & Post Processing

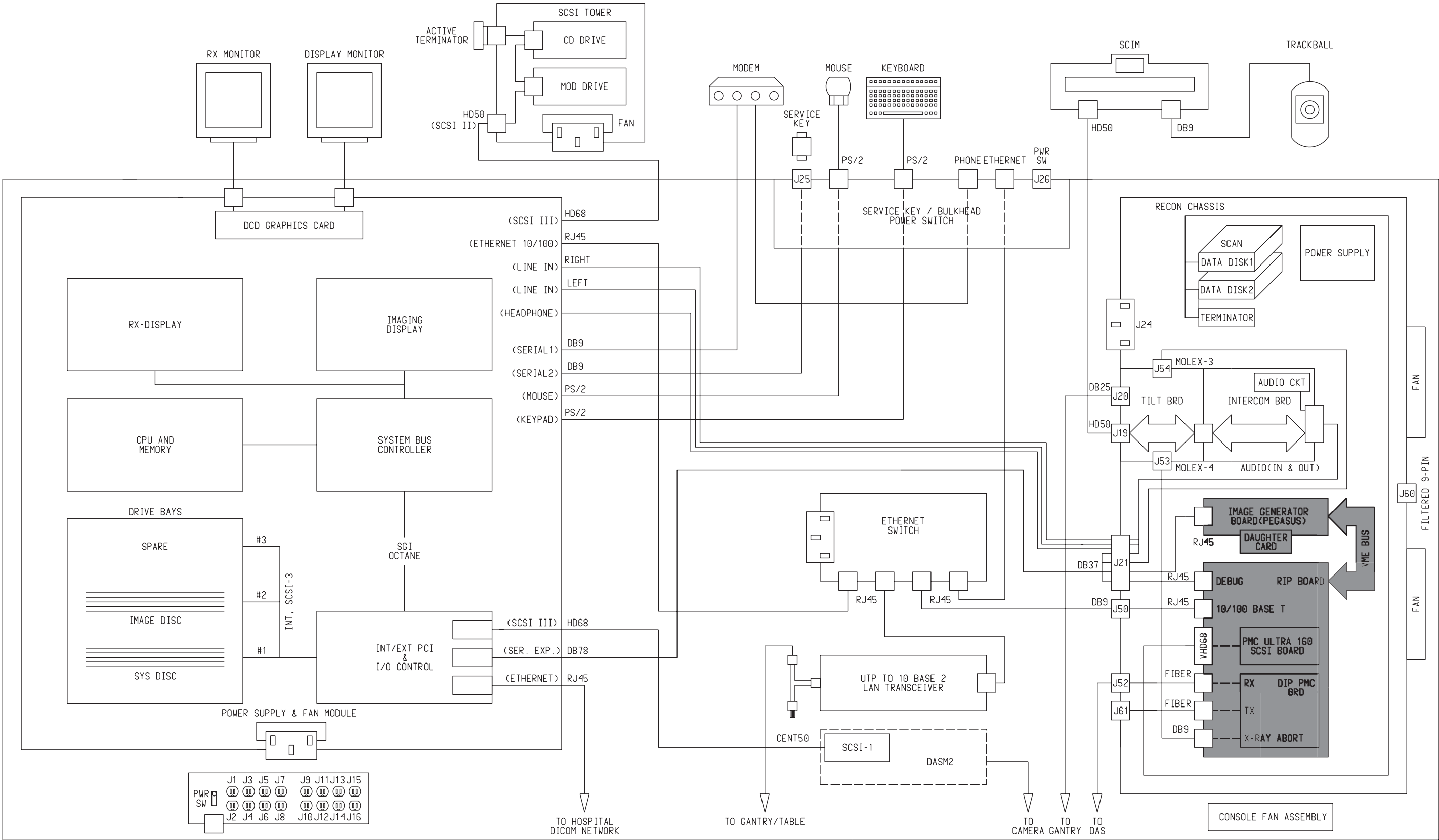


Figure 10-3 Pre-Processing, Filtered Back Projection & Post Processing

Section 4.0 – Image Transfer

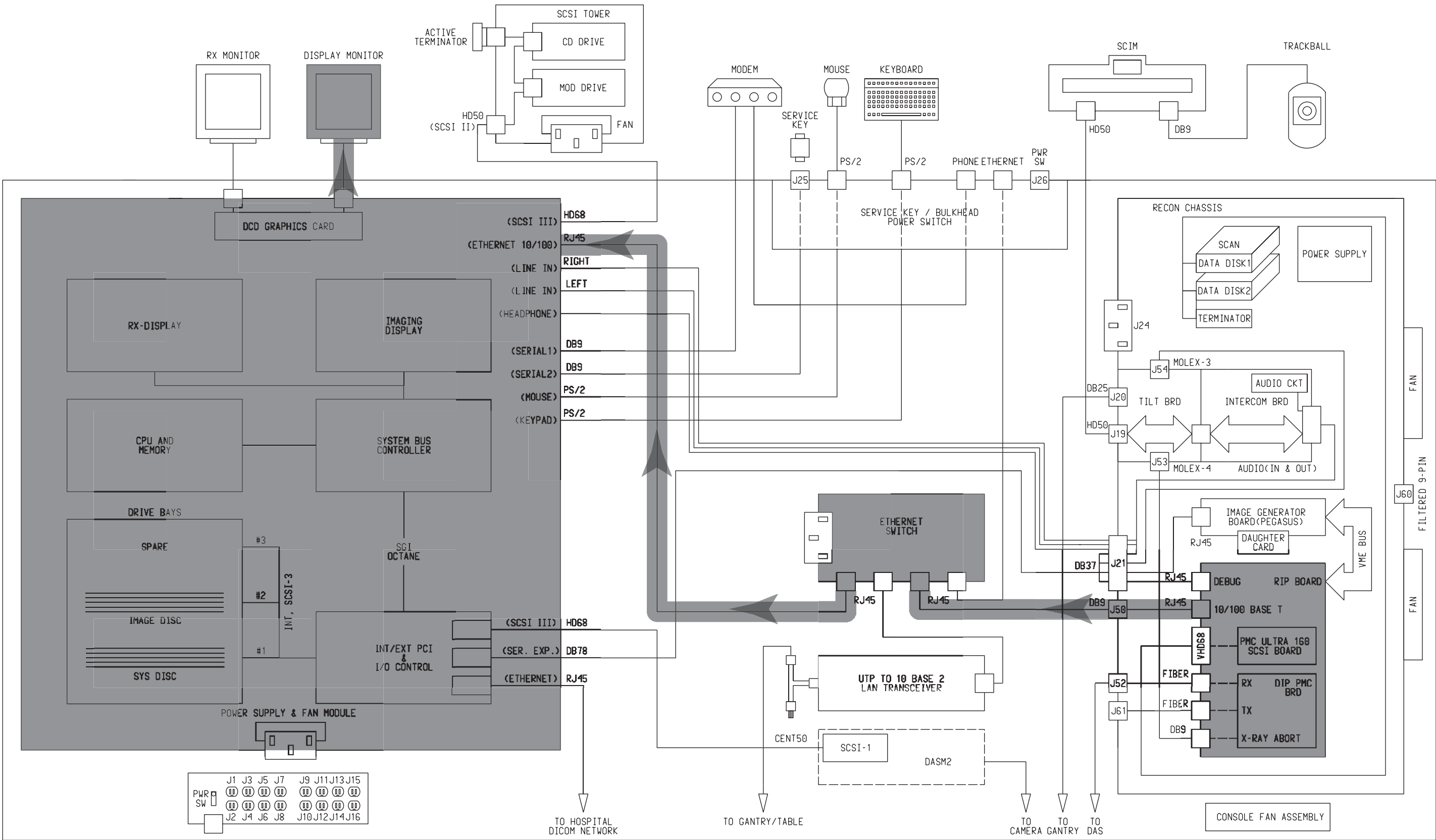


Figure 10-4 Image Transfer

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